NOTES AND REFERENCES

In the following references the bold face numbers refer to the numbers of the literature in the bibliography; p, or, pp, indicates the page or pages; ff, indicates following pages.

In many instances the number of the page is given, but in others, when I refer to a large subject, only the number of a book or paper is indicated, and in such cases the index of the given book should be consulted.

In other cases, when no references are given, and yet the serious and educated reader may occasionally feel perplexed, may I not suggest, in this connection, that wide experience has taught me that we usually forget the structural, not entirely common, subtleties of grammar. We also often ascribe to words a very limited, personal, and *habitual* range of meanings, and, so, some purely linguistic difficulties appear as mysterious 'scientific' difficulties, which they are not. The reader, on such occasions, will be surprised to find what an enormous amount of knowledge may be found in a mature occasional perusal of a good grammar or dictionary, the neglect of which acts as a psycho-logical blockage to the understanding.

CHAPTER II

1—**25.** 2—**212.** 3—**591,** pp. 93, 94.

CHAPTER III

1—The need of International Languages, or a *Universal Language* besides mathematics is becoming increasingly urgent. At present there are several such languages, and in many large cities there are organizations, usually called International Auxiliary Language Associations, with addresses listed in telephone directories. Any of these organizations will gladly supply information about the whole of the international linguistic movement. There is also a number of books written on this subject to be found in the larger public or university libraries. Informations about the *Basic English* of Ogden as a *Universal Language,* consisting of the astonishingly small number of 850 words, which do the work of about 20,000 words, may be obtained from the Orthological Institute, 10 King's Parade, Cambridge, England (see also **376, 377**). In my opinion, the possibilities of the Basic for a scientific civilization are unlimited, *provided* the Basic is *revised* from a non-aristotelian, non-identity, point of view.

The general and serious defect of all of these languages is, that their authors have, as yet, entirely disregarded the non-aristotelian problems of non-identity, and so of *structure,* without which *general sanity,* or the elimination of *delusional worlds* is *entirely impossible.*

CHAPTER IV

1—**590.** 2—**579,** Vol. II, Part IV, *150 ff.; **455-457.** 3—**590.** 4—**457,** p. 249.

CHAPTER VII

1—**317, 318, 319.** 2—**83.**

CHAPTER VIII

1—**92,** pp. 50-52. 2—**91, 92.** 3—**92,** pp. 114, 119, 123, 242. 4—**564, 560,** pp. 16 ff., 28 ff.

CHAPTER IX

1—**487.** 2—**304, 7,** Vol. II, p. 461 ff. 3—**214.** 4—**7,** Vol. II, p. 457. 5—**214,** p. 210. 6—**7,** Vol. II, p. 458. 7—**7,** Vol. II, p. 251. 8—**7,** Vol. II, p. 634. 9—**7,** Vol. II, p. 944. 10—**310,** Chap. V. 11—**364, 365-369, 540.** 12—**9, 10, 12, 13, 196, 210, 211, 370.** 13—**257, 272, 328, 345.** 14—**7,** Vol. II, p. 917 ff., **273, 313, 416.** 15—**7,** Vol. II, p. 690. 16—**7,** Vol. II, p. 961 ff. 17—**7,** Vol. II, p. 644 ff. 18—**7,** Vol. II, p. 803 ff. 19—**7,** Vol. II, p. 869 ff. 20—**363.** 21—**499, 500-504, 127, 128.** 22—**201, 533.** 23—**49.** 24—**7,** Vol. II, p. 59 ff.

CHAPTER X

1—**306**, p. 72 ff. 2—**306**, p. 72 ff. 3—**306**, p. 72 ff., p. 193. 4—**310**, p. 95 ff. 5—**309**, p. 197. 6—**309**, p. 202. 7—**309**, pp. 204, 211. 8—**411**, p. 233. 9—**221**, p. 139. 10—**221**, pp. 144 145. 11—**343**, **344**. 12—**559**, p. 15. 13—**242**.

CHAPTER XI

1—**452**, **453-457**. 2—**82**, **264**, **269**. 3—**470**. 4—**176-184**.

CHAPTER XII

1—**220**, p. 60 (5th edition). 2—**222**, pp. 3-5. 3—**411**, p. 8. 4—**300**, and in a personal letter. 5—**220**, p. 334. 6—**220**, p. 68. 7—**411**, p. 210. 8—**566**.

CHAPTER XIII

1—**452**, **453**.

CHAPTER XIV

1—**264**. 2—**203**, p. 39.

CHAPTER XV

1—**452**, **453-457**. 2—**457**, p. 22 ff.

CHAPTER XVI

1—**449**.

CHAPTER XVII

1—**547**, p. 102. 2—**208**, p. 811 (17th edition). 3—**208**, p. 815 (17th edition).

CHAPTER XVIII

1—**455**, p. 18. 2—**126**, pp. 14, 16. 3—**74**, p. 467, **468**.

CHAPTER XIX

1—**264**. 2—**430**. 3—**468**, p. 133. 4—**468**, p. 141. 5—**264**, p. 153 ff. 6—**264**, p. 153 ff. 7—**222**, p. 166 ff. 8—**289**. 9—**350**. 10—**318**, **319**.

CHAPTER XX

1—**221**, p. 277. 2—**7**, Vol. II, pp. 59-92, **121**, **436**. 3—**216**.

CHAPTER XXI

1—**394**, **395**. 2—Rudyard Kipling's *Verse*. Inclusive edition, 1885-1926. New York. 3—**501**. 4—**221**, p. 197. 5—**552**, **553-557**. 6—**250**, p. 194.

CHAPTER XXII

1—**92**, p. 112. 2—**221**. 3—**221**, pp. 86, 87, **91**, Chap. XIII. 4—**306**, p. 48 ff. 5—**306**, pp. 37-40. 6—**306**, pp. 85, 86. 7—**306**, p. 61-63. 8—**304**, p. 45. 9—**487**, p. 393. 10—**222**, pp. 213, 214. 11—**222**, pp. 262, 263. 12—**394**, pp. 47, 388. 13—**278**, pp. 90-92. 14—**394**, p. 99.

CHAPTER XXIII

1—**394**, pp. 88-94. 2—**394**, pp. 310, 311. 3—**394**, pp. 310, 311. 4—**394**, pp. 403, 404.

CHAPTER XXIV

1—**34**.

CHAPTER XXV

1—All anthropology gives ample evidence. In the present very brief bibliography, consult **172**, **173**, **200**, **298**, **299**, **331-336**, **492**.

CHAPTER XXVII

1—**579** (1st edition), Vol. I, p. 40. 2—**590**, No. 3, 332. 3—**97**. 4—**452**, **453-457**, **579**, and all modern works on 'logic' and the foundations of mathematics. 5—**579**, pp. 63, 65 ff., Vol. I (1st edition).

CHAPTER XXVIII

1—**247.** 2—the whole of primitive and modern mythologies, consult **172, 173, 200, 298, 299, 331-336, 478, 479, 492.** The literature is very large, and cannot be given here. Consult also standard treatises on comparative religion, history of religion, and works of psychiatrists who deal with these aspects of 'mental' ills.

CHAPTER XXX

1—**241,** p. 37. 2—**241,** Fig. 2, p. 94, Fig. 3, p. 140. 3—**241,** pp. 140-142. 4—**42,** pp. 1, 19, 23, 118-120. 5—**42,** p. 23. 6—**99,** p. 897. 7—**34,** p. 230. 8—**34,** p. 180. 9—**34,** p. 233. 10—**34,** p. 242. 11—**411,** p. 237. 12—**411,** p. 10. 13—**15, 16, 286.** 14—**298,** Chap. IX. 15—**181,** Vol. IV, p. 136, **492.** 16—**70.** 17—**487,** pp. 1363, 1334. 18—**487,** pp. 1327-29, 1363. 19—**34,** p. 117. 20—**34,** p. 91. 21—**70.**

CHAPTER XXXI

1—Science. Jan. 22, 1932. 2—**56.** 3—**105, 290, 329, 413, 467, 521, 532,** the literature on this subject is very large, and I give here only examples, see also **110.**

CHAPTER XXXII

1—**72.** 2—**381, 510, 585.** 3—**220,** p. 368 (5th edition).

CHAPTER XXXIII

1—**547,** p. 104. 2—**55,** pp. 89, 174, 221. 3—**411,** p. 191.

CHAPTER XXXIV

1—**147,** p. 79. 2—Most of the above presentation follows **45.** 3—**204,** Vol. II, p. 342. 4—**45,** pp. 260, 261. 5—see, however, **472** and **508.** 6—see C. Runge, *Vector Analysis* (London, New York), p. 178. 7—**148,** p. 49. 8—**147,** pp. 86-88. 9—**45,** pp. 18, 22, 24, 25, 36. 10—**411,** pp. 24, 50, 91.

CHAPTER XXXV

1—**45,** p. 132. 2—**547,** p. 92. 3—**204,** Vol. I, pp. 247, 248. 4—**45,** pp. 158, 159. 5—**547,** p. 66. 6—**547,** p. 91. 7—**457,** pp. 21, 22.

CHAPTER XXXVIII

1—**264,** p. 327 ff., see, however, **348.** 2—**314,** p. 75 ff. 3—**147,** p. 58 ff. 4—**45,** pp. 196, 197, 213.

CHAPTER XXXIX

1—**148,** p. 23. 2—**45,** p. 232. 3—**148,** p. 136. 4—**147,** pp. 177, 178. 5—**547,** pp. 284, 285. 6—**147,** p. 178. 7—**147,** p. 149.

CHAPTER XL

1—**480,** pp. 65, 66. 2—**31,** pp. 17-19. 3—**480,** pp. 467, 468. 4—**32,** p. 38.

CHAPTER XLI

1—**142,** p. 5. 2—**204,** Vol. I, p. 79 ff. 3—**480,** pp. 194-202. 4—**32,** p. 64; **46,** p. 74. 5—**123, 39, 522.** 6—**123,** Vol. III, p. VI. 7—**146.** 8—**457,** pp. 42-44. 9—**32,** later Dirac altered his procedure, see **142.** 10—see last section of appendix 7 in the German, 4th edition of **480,** and also **481.** 11—**32,** p. 185, **63, 64, 65.** 12—**27,** p. 16 ff. 13—**32,** pp. 148, 149. 14—**32,** p. 137; **27,** pp. 25, 26. 15—**27,** pp. 64-67. 16—**27,** p. 38.

SUPPLEMENT III

1—for literature, see under respective names. 2—**249.**

BIBLIOGRAPHY

A complete bibliography for a *non-aristotelian system* would require many volumes and is, therefore, impossible here. The formulation of a *non-aristotelian system,* with the number of scientific facts known in 1933, turned out to be an extremely laborious process.

A non-aristotelian language and attitude differ considerably from the older languages and attitudes, and so a first non-aristotelian system has no literature which would deal directly with the subject. The statement 'that everything has already been said' is, unfortunately, largely true. This introduces serious complexities, because extremely few men have the genius of a Poincaré, and fully realize that the language used in making a statement plays an overwhelming role as to the consequences which eventually follow. If we even grant that 'every thing was said', I must add, 'but not so', and this *prevented the building* of a non-aristotelian system for more than two thousand years. One of the human tragedies can be found in the fact that *wise epigrams* do not work. It takes *a system* which often expresses similar notions; but they must be expressed in a *unified language* of different structure to make them workable.

In giving this extremely abbreviated, insufficient, and, perhaps, even poorly selected bibliography, I had, in the main, three aims: (1) to acknowledge some of my direct obligations; (2) to give to the future student an outline of the type of literature in existence which has bearing on my subject; and (3) to list such books and articles which give further literature.

I was particularly careful to list as few scientific periodicals as possible, because specialists in a given field do not need them; and laymen do not want them. In a number of instances, I have listed only one or two of the latest papers of an author, which give his previous titles.

Because of the lack of linguistic co-ordination, in most cases, I have had few or no opportunities to refer directly to many authors, although the titles of the books usually suggest the material needed. For further data on a given subject, the reader is referred to the respective indexes. As a rule I had to express many similar notions, but from a different angle, and in a different language.

I have prefaced the books, parts, and chapters with many important quotations, only to show that all modern science requires a fundamental non-aristotelian revision. The attentive reader will discover that, although I am in general sympathy with these quotations, yet, in many instances, I would have to express them differently.

In the case of the two volumes of *Colloidal Chemistry* edited by Alexander, which is a collection of important contributions by different authors, I would have to list about one hundred and twenty more titles, and so I mostly refer to the page without giving the name of the author or the title of his contribution; for which I apologize. In a number of instances I have utilized the material given by *Science Service,* as printed in *Science* and I indicate such references by inserting *SS* before giving the date of the issue of *Science.*

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INDEX

In the case of terms of frequent occurrence, only those occurrences are indexed which elucidate the meanings.

A. *See* Aristotelian system, and Trilogies.
$\bar{A}$. *See* Non-aristotelian system and Trilogies.
Abbreviations of terms, 15 f.
Absolute, theory of, 640.
Abstracting, 165 f., 178 f., 235 ff., 371 ff., 426, 507.
  consciousness of, 36 f., 110, 297, 304, 310, 337, 340, 355, 364, 395, 409 ff., 415 f., 424, 426, 434 f., 437, 440 f., 443 f., 449, 470, 480, 489 f., 493, 499 ff., 514, 526, 539, 545 ff., 753 ff.
  neurological function of. *See* Neurological.
  process of, 180, 224, 317, 379 ff., 482, 694.
Abstractions, orders of, 42, 56, 94, 105, 107 f., 179 f., 183, 230, 232, 240, 254, 262, 272, 277, 279 f., 288, 290 ff., 306 ff., 331 ff., 372, 380 ff., 389 ff., 399, 402, 409, 425 ff., 454 ff., 474 f., 478, 507, 513, 549, 552 ff., 575 f., 578, 583, 613, 637, 645, 647, 655 f., 664, 671, 679 f., 694, 716, 718, 754 f.
  confusion of orders of, 184 f., 216, 278, 294, 304, 320, 327, 336, 363, 405, 417 ff., 423, 432, 443 ff., 469, 502, 514, 637, 649.
  difference between orders of. *See* Translation.
  of man and animals, 394, 500.
  mathematical, 67 ff.
  physical and daily, 67 ff.
Achilles and the tortoise, 71, 247.
Ackermann, 767, 773.
Acquisitiveness, 519, 521.
Actinians, experiments with, 346.
Action, 679 ff., 688, 724.
  by contact, 235 ff., 637 ff., 657.
  least, 408, 589, 680.
Adams, 767.
Additivity, 89, 172, 198, 224, 264 f., 412 ff., 527, 603 ff. *See also* Non-additivity.
Adjustment, 29, 184, 224 f., 239 ff., 270 ff., 294, 316, 355, 474, 532, 574, 613, 632, 647, 676, 693, 701, 727.
Adler, 767.
Adrenal glands, 128. *See also* Endocrine glands.
Affects (affective), configurations, 23.
  components, 50, 279, 357, 419.
  disturbances, 21, 305.
  evaluation, 22, 302.
  meanings, 27.
  responses, 268, 329, 501, 716, 728.
Aggregates, theory of, 69, 208, 280 ff., 419, 428.
Agreement, 201, 274, 287, 364, 375, 419, 549, 619, 761.
Aleph, cantorian, 208.
Alexander, 767, 769.
Algebra of states and observables, 69.
'Allness', 379, 404, 455 ff., 469, 471 ff., 492, 545. *See also* Non-all.
Amoeba, experiments with, 345 f.
Analysis situs, 281, 658.
Anglo-Saxon philosophy of selfishness, 521.
Animals, copying of. *See* Copying.
  experiments with, 117. *See also* under names of animals.
  reactions of, 51.
  and man. *See* Man.
Antagonistic muscles, 349.
  processes, 352, 362.
Anthropology, 38 ff., 48, 377, 400, 401, 514, 752, 764.
  British School of, 400.
  French School of, 400, 401.
  Polish School of, 400, 401.
Anthropometer. *See* Structural Differential.
Anthropomorphism, 89, 200, 211, 343 f., 347, 351, 385, 727.
Anrep, 315, 767.
Apes, decorticated, 177. *See also* Cortex and Thalamus.
Aphasia, 19 f., 44, 131, 183 f.
  semantic, 19 f., 44.
Apple, 96, 384, 402, 456, 464 f., 758.
Approximation, 596 f., 611 ff.
Archimedes, 505.
Aristotle, 43 f., 85 ff., 90 ff., 101, 203, 251, 288, 302, 367, 371, 726.
Aristotelian system, 7 ff., 27, 45, 61, 93 f., 130, 141, 145, 156, 165, 168, 187 f., 192, 203, 233, 273 f., 287, 303, 371, 401, 404 f., 408, 411, 432 f., 447, 452, 473, 481, 537 ff., 638, 663, 726.
Arithmetics, 251, 265.

Arrays, 205, 208 ff., 711.
Arrested development, 193, 200, 296, 305, 321, 332 f., 337, 378, 495 ff., 501, 503, 508, 515.
Ascidians, experiments with, 346.
Associations and the process of relating, 284, 289, 512, 613.
Assumptions, 16, 280, 441, 504.
Atom, 113, 283, 685 ff.
Atomic processes, 99, 322, 563, 688, 693, 721, 736. *See also* Electrons.
Auditory types, 460.
Autoerotism, 494, 496, 513. *See also* Sex disturbances.

Babcock, 768.
Bacon, F., 541.
Bacon, R., 768.
Bagg, 768.
'Bahnung', 299.
Baker, 567.
Baldwin, 33, 768.
Bancroft, 511, 768.
Baron, 768, 772.
Barrow, 574.
Basic English, 763.
Bayliss, 768.
Becker, 768.
Begging the question, 403, 443, 496.
Behaviourism (ists), 71, 74, 303, 334, 359.
Bell, i, 2, 3, 55, 85, 133, 188, 204, 245, 247, 367, 469, 569, 571, 635, 768, 781.
Bent, 768.
Bentley, 20, 563, 768.
Bergson, 244, 737, 744.
Beri-beri, 126 f.
Bernstein, 768.
Berry, 743.
Biggs, 569, 721, 768.
Biology, non-aristotelian, 30, 124, 345, 644.
Bird, 768.
Birds, experiments with, 36, 183, 301.
Birkhoff, 768.
Birtwistle, 569, 702, 725, 768.
Bleuler, 323, 515, 523, 768.
Bliss, 768.
Blockage, semantic, 18, 147, 193, 205, 227, 248, 263, 299, 305, 327, 437, 440, 443, 456, 476, 484, 490, 538, 658, 701, 719.
Boas, 768.
Bôcher, 569, 768.
'Body-mind' problem, 111, 119, 380, 498, 512, 532, 542, 545.
Bohr, 283, 690, 717, 723, 725 f.
Bolton, J. S., 320 f., 510, 768.
Bolton, L., 768.
Boltzmann, 210.
Bolyai, 86, 505.
Bolzano, 205.
Bonola, 768.
Boole, 85, 210, 310, 768.
Borel, 567.
Born, 569, 637, 671, 673, 714, 717, 768.
Bose, 724.
Botcharsky, 768.
Bousquet, 768, 776.
Bovie, 120, 768.
Bradley, 741.
Bragg, 768.
Brain. *See* Cortex and Thalamus.
  diseases, 510.
Bréal, 19.
Breitwieser, 768.
Bridges, 767, 769, 776, 781.
Bridgman, 2, 3, 368, 542, 569, 648, 769.
Brigham, 769.
Brill, 769.
Brillouin, 769.
Broad, 769.
Broglie, de, 718, 720, 724, 769.
Brouwer, 64, 79, 94, 204, 249, 747 f., 769.
Brunot, 769.
Brunschvicg, 769.
Burrow, 521, 526, 769.
Butterflies, experiments with, 124.

Cajori, 769.
Calculus, 490, 576.
  absolute, 568.
  differential, 213, 215, 287, 292 ff., 324, 543 ff., 597, 619 ff., 640, 694.
  extensional, 711.
  infinitesimal, 215.
  integral, 213, 215, 592 ff., 612, 622 ff., 640, 694.
  matrix, 706 ff., 717.
  non-aristotelian, semantic reactions of, 602.
  operator, 608, 711 ff.
  structural, 253, 259.
  tensor, 617 ff., 658.
  vector, 354, 609 ff., 617 ff., 710 ff.
Campbell, 769.
Cannon, 769.
Cantor, 93, 205, 209, 565, 756, 769.
Capitalism, 270 ff.
Carmichael, 85, 144, 368, 729 ff., 769.
Carnap, 769.
Carrel, 769.
Carroll, 76, 368, 452.
Cassirer, 769.
Castillon, 152.
Cataphoresis, 115.
Caterpillars, 167, 239.
Cathode rays, 118, 686.
Cats, 2, 339, 342, 511.

Causality, 214 ff., 612 ff. *See also* Non-causality.
Cause-effect, 93 f., 126, 214 ff., 310 f., 523.
Cayley, 710.
Cerebral localization, 630 ff. *See also* Cortex and Thalamus.
Chemical elements, 685.
Chickens, experiments with, 605.
Child, C. M., 30, 53, 99, 101 ff., 124, 188, 235, 341, 345, 541, 644, 769.
Chromosomes, 117.
Church, Alonzo, 433, 769 f.
Church, Archibald, 770.
Chwistek, 20, 204, 541, 748, 753, 770.
Cicna intestinalis, experiments with, 346.
Circularity of knowledge, 220 ff., 398 ff., 471 ff., 759.
Civilization, 62, 201, 291, 344, 400, 473, 486, 508, 514, 533, 553, 727, 756.
  suggestions for, 555 ff.
Clark, 770.
Class, 131, 208, 247, 255, 432, 742 ff.
Coghill, 326, 341, 770.
Cohen, E., 770.
Cohen, M. R., 770.
Collins, J., 516, 770.
Collins, J. L., 768, 770.
Colloidal behaviour, 8, 111 ff., 193, 322 f., 401, 508, 511 f., 531 f., 541 f.
  disturbances and injury, 200, 512.
Colour-blindness, 390.
Combinations of higher order, 340, 462.
Commercialism, 42, 83 f., 270, 295 f., 305, 508, 517, 519, 521, 549, 555 ff., 558, 756.
Compactness. *See* Density.
Complex, Jehovah, 153.
  Oedipus, 501.
Complexes, 361, 502, 512, 527 ff., 535.
Compton, 244, 692, 702.
  effect, 244.
  theory, 702.
Compulsory actions, 338, 368, 546.
Conditional reflexes, 19, 315 ff., 326 ff., 353, 512.
Conditionality, degrees of, 36, 195, 327, 331 ff., 335, 337 ff., 341, 343 f., 351, 443, 546 f.
Conflict, nervous, 529 f.
Confucius, 38.
Conklin, 770.
Consciousness, 3, 412 ff., 440, 492 f., 503.
  of abstracting. *See* Abstracting and Abstractions.
Conservation of energy, 212, 232, 656, 679, 720, 724.
Contact, action by. *See* Action.
Continuity, 151, 212 f., 249, 287, 578, 582, 612, 640, 725. *See also* Discontinuity.
Cook, 770.
Coolidge, 770.
Co-ordinates, 139, 242, 283, 573, 576, 583, 586, 589, 597, 616, 619 ff., 646, 651, 661 ff., 679, 702, 734.
Copernicus, 550, 692.
Copying animals in nervous reactions, 7 f., 36, 62, 73, 200, 231, 301, 305, 327 ff., 321, 356 f., 363, 378, 397, 410, 481, 484, 508, 538, 543, 547, 654.
Corpus Errorum Biologicorum, 5.
Correct symbolism, 80, 84, 219, 530.
Correspondence, one-to-one, 57, 209 ff., 258, 423, 462 ff.
Cortex, cerebral, 157, 177 f., 188, 193, 268, 289 f., 299, 315, 320, 333, 338 ff., 349 ff., 362, 369, 412, 422, 425, 456, 468 f., 510 f., 758.
Cortical influence, dynamogenic, 350, 352.
Cosmic rays, 117, 237.
Coulomb, 646.
Courant, 770.
Couturat, 152, 367, 770.
Creative 'thought', 537, 565, 676.
Cretinism, 128, 523.
Crile, 770.
Crime, criminals, 269, 305, 445, 524 f.
Crookshank, 491, 542, 770.
Cullis, 770.
Culture, degree of, 377, 439, 490, 553.
Cunningham, 770.
Curvature, 590 ff., 645, 679 ff.
  of space-time, 233 f., 243, 318, 384, 386.

Daltonism, 390.
Dantzig, 256 f., 770.
Darrow, C. W., 770.
Darrow, K. K., 770.
Darwin, C., 271, 550.
Darwin, C. G., 770.
Date. *See* Space-time and 'Time'.
Davenport, 770.
Davis, H. T., 770.
Davis, T. L., 770.
Day, 66.
Dedekind, 205, 770.
Deduction, 69, 308.
Degree. *See* Kind.
Déjerine, 770.
Delacroix, 770.
Delayed action, 317, 319, 342, 349, 355, 357, 362, 422, 424, 446, 468.
  reflexes, 330, 360.
Delusions, 61, 77, 79, 81, 87, 137, 198, 298, 408, 418, 422 ff., 428, 434, 440, 456, 460, 472, 475, 481 f., 492 f., 512, 550, 553, 637, 697, 699, 716. *See also* Hallucinations and Illusions.
Delusional values. *See* Values.
Demand, supply and, 521, 548.
Dementia praecox. *See* Schizophrenia.

Democracy, 486.
Democritus, 685.
Demons, 89, 105, 201, 347 ff., 514.
Density, 213, 249, 680.
Denumerability, 209.
Dercum, 770.
Descartes, 134, 146, 251, 329.
Description and inference, confusion of, 753, 755. *See also* Identification.
Descriptive level, 217, 359 f., 392 ff., 444, 478, 753, 755.
Determinants, 266, 706 ff.
Determinism, 94, 217, 221 f., 310, 535, 550 ff., 760.
Development, arrested, 193, 200, 296, 305, 321, 332 f., 337, 378, 495 ff., 501, 503, 508, 515.
  stages of, 494.
Dewey, 771.
Dickson, 771.
Differences, 514, 517.
  vertical and horizontal, 395 ff. *See also* Stratification.
Differential activation, 350.
  calculus. *See* Calculus.
  geometry, 233, 575, 642, 658.
  methods, 93, 602.
  Structural. *See* Structural Differential.
Dimensionality, 93, 151, 234, 619 ff., 626 ff., 661 f., 665 ff. *See also* Four-dimensional.
Dinosaurs, analogy of the, 72.
Diplomacy, 521.
Dirac, 322, 699, 714, 717, 765, 771.
Disagreement, origin of, 82.
Discourse, universe of, 225, 254.
Discontinuity, 249, 688, 725 f. *See also* Continuity.
Disinhibition, 353. *See also* 'Inhibition'.
Doctrinal function, 94, 144 ff., 437, 534.
Dog, experiments with, 177, 183, 326, 329 ff., 333, 358 ff., 511 ff.
Dostal, 569.
Doubt, 439 f.
Dresden, 771.
Drobisch, 771.
Dynamic and static. *See* Translation.
Dynamic gradient, 101 ff., 108, 235, 345, 348, 541.
Dynamics, 647, 714.

E. *See* Euclidean and Trilogies.
$\bar{E}$. *See* Non-euclidean and Trilogies.
Earthworm, experiments with, 124 f.
Eckart, 714, 771, 781.
Economics, 42, 49, 76 f., 84, 270 f., 273 f., 295 f., 305, 508 f., 517 ff., 548 f., 554 ff., 756, 761.
Eddington, 53, 85, 151, 218, 223, 234, 245, 368, 565, 569, 625, 654, 666, 668, 678, 680, 683. 685, 771, 781.
Edinger, 157, 177.
Education (al), 45, 47, 63, 88, 182, 185, 187, 199, 218, 243, 295 f., 299, 304 ff., 320 f., 327 ff., 334, 343, 356, 364, 379, 399, 411, 413 ff., 422, 429 ff., 440, 447, 459, 466 f., 481 ff., 489 ff., 500, 511, 543, 552, 554, 560, 567, 575, 604, 760.
Einstein, 2, 7, 30, 57, 60, 64, 66, 80, 86, 89, 91, 106, 148, 162, 214, 223, 243, 265, 268, 286, 299, 318, 344, 347, 381, 386, 488, 513, 541, 554, 565, 571, 575 f., 619, 622, 630, 635, 640, 642, 646, 648, 653 ff., 666 ff., 672 ff., 685 f., 689, 698 f., 718 ff., 724, 726, 733, 735, 771, 775.
Einstein theory, 7, 30, 64, 97, 107, 229, 248, 263 ff., 275, 289, 487, 569, 576, 604 ff., 612, 618, 626, 635, 640, 648 ff., 656, 663, 668 ff., 712, 718 f., 726.
el, elementalistic. *See* Elementalism.
Electron, 108, 163, 322, 387, 612, 686, 690 ff., 696, 700 ff., 714 ff., 720 ff.
Electricity, 364.
  and colloidal behaviour, 508.
Electrical phenomena in living tissue, 115 f.
Electrochemistry, 685.
Electromagnetic phenomena, 386, 700.
  theory of the structure of 'matter', 687.
  waves, 236 f.
Electromagnetism, Maxwell theory of, 642, 644, 663, 673, 686, 695 f.
Elementalism, 31, 64, 87, 93, 107 f., 189, 192, 243, 265, 303, 371, 379, 394, 407 f., 433, 455, 492, 539, 542, 551, 653, 699, 747 ff. *See also* Non-elementalism.
Ellis, 771.
Empiricism, 1, 87.
Endocrine disturbances, 128, 401. *See also* under names of glands.
  glands, 331, 522. *See also* under names of glands.
Energy, 679, 702, 733 ff.
  law of conservation of, 212, 232, 656, 679, 720, 724.
Enriques, 771.
Entropy, 212, 232, 680, 760.
Environment, 130, 168, 175, 198, 270, 346.
Epilepsy, 118 f.
Epimenides, 435 f., 739.
Epistemology, 65, 78, 101 ff., 139, 173, 204, 218, 248, 256, 313, 327, 394, 440, 540, 543, 547, 554, 568, 573, 640, 698 f.

Epithets, question begging, 403, 446, 496.
Equations, degrees of, 172, 266, 585, 595.
differential, 265, 595 ff., 637, 646, 724.
field, 642.
functional, 608.
hamiltonian, 714 f., 722.
as language of relations, 353.
matrix, 707 ff.
non-linear, 265.
tensor, 707 ff.
Equivalence, 209.
principle of, 735.
Error, Gauss' curve of, 725.
Essences, 674.
Ether, 674.
Ethics, 550 f.
Eucharis, experiments with, 348.
Euclid, 27, 90, 205, 441 ff., 504 ff., 641 f., 645.
Euclidean geometry, 27, 85 f., 95, 232, 265, 405, 441, 611, 615, 622, 646, 667 ff., 674, 756.
'space', 615.
Euler, 589.
Europe, confederated, 522.
Evaluation, 22, 31, 164, 170, 189, 192, 194 ff., 202 f., 228, 247, 264, 275, 294, 298, 301, 304, 307 f., 320, 329, 338, 355, 375, 378, 382, 396, 406 ff., 421, 425, 428, 435, 440 f., 443, 448, 454, 456 f., 460, 466 ff., 478, 480, 488, 492, 498, 500, 502, 506, 509 f., 512, 517 f., 528, 530, 532, 539, 545, 554 ff., 558 f., 755, 761.
infantile. *See* Infantile.
semantic factors in, 29.
Events, 219, 664, 667 ff., 677, 755.
recognition of, 386.
Event-process, four-dimensional, 387, 464.
Events, language of, 224, 243, 386. *See also* Point-event and World-line.
Excitation, degrees of, 350, 353, 360, 362.
negative, 351 ff., 355, 361, 364.
Excluded third, law of, 281, 404 f., 430 f., 747 ff., 760.
Exhibitionism, 520 f. *See also* Infantilism and Sex disturbances.
Existence, 79 ff., 93, 141.
Expectations, semantic shocks from unjustified, 472.
Experience, 278, 294, 299, 375 f., 394, 456, 523, 666, 754.
Exponential rules, 264, 354 f.
Extension, 15, 94, 135, 141, 152, 170 ff., 176, 179, 182, 208, 219, 286, 381, 632, 639, 711, 724, 726, 746. *See also* Intension.
Extensional punctuation. *See* Non-aristotelian.
Extra-neural means, 109, 231, 240, 291 f., 298, 300, 375 f., 395.
Extroversion, 87 ff., 174 ff., 425, 461, 485. *See also* Introversion.
Fajans, 771.
Falsehood, neurology of, 529 f.
Fan, rotating, experiment with, 382 f., 472, 694.
Faraday, 643 f., 646, 700.
Ferenczi, 771.
Fermat, 722.
Fertilization, artificial, 117, 124.
'Fido', 8, 167, 390 ff., 409 f., 413 ff., 458, 503, 538 ff., 545, 549.
Findlay, 112.
Finite, variable (Cantor), 93, 207, 756, 759.
size of the world, 228 f.
Fischer, B., 177.
Fischer, M. H., 7, 491, 771.
Fisher, 771.
Fishes, experiments with, 116, 178, 338 f., 605.
FitzGerald, 674, 677.
Fixation, 497, 547.
Foeringer, 768, 771.
Forgetting, 491 f.
Formalism, 207, 276, 437, 761.
Four-dimensional methods, 93, 139 ff., 263, 459, 536.
theory of language, 94, 243, 513.
world, 234, 293, 630 ff., 665 ff., 726.
Fourier, 601, 715, 723.
Fraenkel, 771.
Frame of references, 242, 616, 729.
Fraud, 83.
Frazer, 771.
Freeman, 511, 771.
Frege, 131, 251, 255, 771.
Freud, 147, 491 ff., 501, 534, 536, 550, 771 f.
Freundlich, E., 772.
Freundlich, H., 115, 772.
Frog, decorticated, 511.
parthenogenic, 124.
Function (al), 93 f., 131, 133 ff., 163, 179, 191, 259, 263, 266, 275, 281, 285 f., 290, 311, 316 f., 328 ff., 333, 379, 412, 539, 574 f., 595, 597 ff., 610 ff., 710, 726, 744 ff.
doctrinal. *See* Doctrinal function.
Funk, 118, 126.
Funktors, 748.

Galileo, 247, 574, 654, 730.
Gamma-rays, 237, 686, 716.
Gantt, 123, 315, 772, 774, 781.
Gauss, 86, 622, 642, ff., 725, 747.
Geddes, 49.
Geodetics, 625, 647, 663.

Geometry, 151, 242, 251, 267, 575, 583, 595, 645.
analytical, 146, 251.
co-ordinate, 615 ff.
differential, 233, 575, 642, 658.
euclidean, 27, 85 f., 95, 232, 265, 405, 441, 611, 615, 622, 646, 667 ff., 756.
four-dimensional, 324, 449, 576, 647, 658, 667 ff.
non-euclidean, 27, 32, 64, 91, 233, 399, 604, 642 f., 658, 669, 701.
of paths, 151.
projective, 265.
riemannian, 573, 615, 640, 646, 667.
Gestalt psychology, 74.
Gibbs, 772, 780.
Glands of internal secretion. *See* Endocrine and under names of glands.
salivary, 330 ff.
Godefroy, 567.
Goitre, 523. *See also* Thyroid.
Gomperz, H., 772.
Gomperz, T., 772.
Gonads, 522.
Gonseth, 772.
Goodspeed, 772.
Goudsmit, 692 ff.
Gradient. *See* Dynamic gradient.
excitation-transmission, 103 f.
physiological, 101, 103, 323.
Graham, 112.
Grassmann, 618.
Graven, 185, 397, 424, 540, 772.
Gravitation, Newton's law of, 626, 629, 635, 646, 656, 680, 700, 735 f.
Great Britain, 522.
Gregariousness, 519.
Gregory, 772.
Groot, 772.
Groups, theory of, 69, 266 ff., 280 ff., 607, 658.
Gurwitch, 768, 772.
Guye, 772.

Haas, 569, 772.
Hahn, von, 118.
Hall, 772.
Halliburton, 772.
Hallucinations, 61, 79, 81, 294, 297, 422 f., 428, 434, 456 f., 460, 637, 699. *See also* Delusions and Illusions.
Halter, 768, 772.
Hamilton, 565, 589, 618, 704, 722.
Hammarberg, 510.
Hance, 772.
Hanson, 772.
Hate, 439 ff.
Hay fever, 128, 331, 375, 415, 546.
Head, H., 20, 22, 341, 772.
Heaviside, 133.
Hedrick, 781.
Heilbrunn, 772.
Heisenberg, 99, 107, 214, 223, 310, 426, 541, 563, 603, 698, 714 ff., 725, 760, 772.
Helmholtz, 644.
Helson, 324, 772.
Henderson, 772.
Herrell, d', 772.
Herrick, 157 ff., 236, 313, 320, 345, 350, 386, 537, 644, 772.
Hertz, 236, 644, 673.
Heterological-autological problem, 739 ff.
Heys, 772.
Hickson, i.
Hilbert, 204, 505, 748, 767, 770, 772 f.
Hinkle, 773.
Hobson, 773.
Hollander, 773.
Holmes, 773.
Holt, 773.
Homosexualism, 513, 524, 526. *See also* Sex disturbances.
Hospitals for the 'mentally' ill, 532 ff.
'Human nature', 33, 40, 45, 107, 182, 228, 322, 329, 441, 446 f., 466, 477, 540, 547, 757.
Hunter, 773.
Huntington, 151, 773, 781.
Huygens, 720, 722.
Hygiene, semantic, 17, 175, 489. *See also* under Semantic.
Hyperpituitarism, 129.
Hypnosis, 491.
Hypopituitarism, 129.
Hysteria, 129, 368, 498.

Idealism, 87.
Identification, 105, 184, 187, 194 ff., 201 ff., 225, 228, 260 f., 263, 265, 278, 294, 296 ff., 302, 304, 327, 338, 379, 384, 396, 399 ff., 403, 405 ff., 417 ff., 428, 434, 440, 452 ff., 458, 460, 462, 470, 483, 491 ff., 506, 514, 516, 538 ff., 545 f., 549 ff., 568, 606, 637, 639, 649, 663 f., 671, 673, 676, 678, 693, 699, 701, 716, 719, 749 ff., 760 ff.
of descriptions and inferences, 417 ff., 448, 479.
Identity, 3, 93, 134, 194 f., 201 ff., 294, 304, 371, 373, 379 ff., 386, 400 ff., 417 ff., 462, 469, 473, 481, 508, 566 f., 638 f., 649, 656, 674 f., 681, 695, 719, 751 ff., 760 ff.
Idiocy, 118, 129, 291, 309, 509 f., 516 ff., 545.
Idiosyncrasies, 498.
Ignorance, 377.
Ill, 'mentally'. *See* 'Insane'.
Illegitimate totalities, 430.

Illness, symptoms of, 118 f.
Illusions, 61, 79, 81, 294, 298, 422 f., 428, 434, 440, 456 f., 460, 637, 699, 716. *See also* Delusions and Hallucinations.
Imbecility, 129, 291, 309, 316, 321, 452, 476, 495, 509 f., 516, 519, 524.
Implication, Lewis' theory of strict, 182.
Indefinitesimal, 205, 218, 580 ff.
Indeterminism, general principle of, 94, 221 f., 310 f., 550. *See also* Uncertainty.
Infantile evaluation, 187, 466 f., 555 ff.
Infantilism, 41, 305, 313, 401, 422, 425, 462, 486, 491 ff., 508 ff., 515 f., 523, 544, 550, 558 f., 757.
Inferences and descriptions, confusion of, 753, 755. *See also* Identification.
Inferential knowledge, 66, 217, 240, 326, 407, 448, 460, 478 ff., 753, 755.
'Infinite velocity', 156, 162, 192, 213, 263, 658, 677.
'Infinite' size of the world, 228 f.
'Infinitesimal', 208, 214 ff., 218, 566, 580.
Infinity (ies), 93 ff., 141, 162, 204 ff., 214 ff., 247, 263, 566, 604, 747 ff., 756.
Infusoria, experiments with, 124.
'Inhibition', 156, 163, 169, 177, 181, 193, 315, 328, 341 ff., 358 ff., 362, 440, 511.
'Insane', 'insanity', 44 ff., 75, 87, 105, 149 f., 169 f., 177, 185 ff., 195 f., 199 f., 225, 269, 288, 295, 298 f., 306 ff., 329, 335, 337, 362, 397, 408, 419 ff., 462, 493 ff., 508 f., 514, 574, 727 f., 750, 760.
  hospitals for, 532 f.
Insects, study of, 339.
Instinct, maternal, 127.
Integration, 188, 230, 279, 484, 504, 526, 532, 574.
Intension, 93, 152, 170 ff., 182, 208, 465. *See also* Extension.
Interdependence, human, 270, 557.
Internal secretion, glands of. *See* Endocrine and names of glands.
International Congress of Mental Hygiene, First, 317.
  congresses, mathematical, 234.
    non-aristotelian, 555, 560.
    physical, 234.
    scientific, 558.
  language, 52.
  Non-aristotelian Library, 554 ff.
    Review, 555.
    Society, 554 ff., 560, 567, 569.
  peace, 17.
  semantic problems, 525 f.
Interval, 32, 234, 625, 668.
Introspection, 359 f.
Introversion, 87 ff., 174 ff., 425, 461, 485.
Invariance, 148, 263, 267, 281, 284 f., 385, 534, 589, 625, 640, 667 ff., 680, 761.
Investigatory reaction, 357.
Irrational numbers, 247, 249, 259, 263.
Irritability, 102, 127, 349. *See also* Protoplasm.
Isotopes, 691.

Jacobi, 722.
James, 773.
Jeffreys, 773.
'Jehovah complex', 77, 153.
Jelliffe, 19, 368 f., 493 ff., 509, 773.
Jelly fish, experiments with, 123.
Jennings, 5, 6, 101, 303, 773.
Jespersen, 773.
Jevons, 340, 749, 773.
Johnson, 773.
Johnstone, 773.
Jones, 773.
Jordan, 714, 717, 724.
Jørgensen, 773.
Jung, 773.
Just, 773.

Kaufmann, 773.
Kempf, 773.
Kennedy, 569, 654, 773.
Kepler, 692.
Keynes, 773.
Keyser, 19, 94, 131, 133, 144 ff., 188, 251, 268, 282, 286 f., 437, 537, 671, 753, 773.
Kind and degree, 254.
Kinder, 773.
Kinematics, three-dimensional, 647.
Kinesthesis, 230, 265, 274, 474.
Kipling, 334, 764.
Kirchhoff, 247, 722.
Klein, 781.
Kline, 773.
Klugh, 773.
Knowledge, content of. *See* Structure.
  inferential. *See* Inferential.
  structure of. *See* Structure.
Knudson, 773.
Koffka, 774.
Köhler, 774.
Kohlrausch, 644.
Kormes, 774.
Korzybski, i, 99, 186, 317, 537, 747 ff., 774. *See also Manhood of Humanity.*
Kretschmer, 774.
Kronecker, 747.
Kupalov, 774.

Labels, 219, 228, 250, 303, 387 ff., 424 ff., 471 ff., 476, 619, 755.
Lagrange, 582, 589.

Lambert, 152.
Lang, 511, 774.
Langdon-Davies, 774.
Langworthy, 774.
Language, 20, 130, 224, 229, 250 ff., 276, 281, 301, 311, 339, 433, 461.
  additive (linear), 172, 264.
  aristotelian, 141, 265, 336, 371.
    structure of, 141.
  arithmetical, 265.
  asymmetrical, 649 f.
  elementalistic, 30, 35, 91, 105 f., 110, 243, 303, 306, 318, 321, 478, 657, 673, 693, 752.
  of events, 243.
  four-dimensional theory of, 93 f., 243, 318, 324, 383, 464.
  human function of, 327 ff.
  ideal, 259.
  international, 52, 763.
  of mathematics, 355, 563, 632.
  non-additive (non-linear), 172.
  non-aristotelian, 450, 484, 727.
  non-elementalistic, 26, 30 ff., 105 f., 136, 289, 305, 318 f., 358, 382 ff., 458, 648, 657.
  of order, 161, 163, 435.
  polysynthetic, 463.
  of relations, 69, 259.
  of schizophrenia, 185.
  of science, 55, 254.
  self-reflexiveness of, 58.
  of 'sensations', 372 f.
  of similarity vs. differences, 254, 264 ff., 382 ff.
  of similar structure, 507, 639, 672, 677, 679, 706.
  structure of. *See* Structure.
  of subject-predicate, 62. *See also* Subject-predicate.
  of tensors, 146. *See also* Tensors.
  three-dimensional, theory of, 93 f., 243, 318.
  of 'true' and 'false', 463.
  universal, 52, 763.
  vector, 242.
Lashley, 295, 774.
Lasswell, 774.
Law, Coulomb's, 646.
  of excluded third. *See* Excluded third.
  of identity, 197, 749 ff.
  of supply and demand, 521, 548.
  Weber's, 238.
Laws of 'nature', structural, 625.
  scientific, 378.
  of 'thought', 197, 210, 310, 404, 747 ff., 760.
  universal, 287, 619, 626, 640, 689.
League of Nations, 522, 526, 537, 558, 560.
Learned, 774, 781.
Least action, principle of, 408, 589, 680.
Leathem, 567, 781.
Leduc, 774.
Leibnitz, 52, 134, 152, 210, 227, 231, 287, 574, 595, 635, 646, 694.
Lenzen, 774.
Leśniewski, 748, 774.
Level, un-speakable. *See* Un-speakable.
Lévy-Bruhl, 514, 774.
Lewis, C. I., 151 f., 182, 774.
Lewis, G. N., 774, 780.
Lewis, N. D. C., 774.
Lie, S., 268.
Lies. *See* Lying.
Light, theories of, 720.
  velocity of, 644, 648 ff., 659 ff., 674, 678, 730 ff.
Lillie, 114, 774.
Limits, theory of, 566.
Linearity, 89, 172, 264, 266, 268, 527, 586, 603 ff., 612, 710.
Lobatchevski, 86, 441 ff., 643, 645, 774.
Locke, 328.
Loeb, 30, 102, 124, 313, 346 f., 541, 554, 774 f.
'Logic', 70 ff., 90, 93, 95, 141, 168, 200, 204, 210, 215 ff., 220, 244, 251, 253, 281, 304, 310, 371, 401, 404 f., 457, 461 f., 537 ff.. 551, 559, 699, 726, 749, 752, 759 f., 764.
  non-chrysippian, 92, 747 ff.
Long, 775.
Lorentz, 566, 657, 666 ff., 673 ff., 686, 721, 733, 775.
Lorentz - Einstein transformation, 566, 654 f., 657, 660 f., 666, 668, 670, 673, 679, 721, 733.
Lotka, 775.
Loucks, 781.
Love, 520, 522.
Lucas, 775.
Lucretius, 204.
Łukasiewicz, 93, 210, 281, 310, 461, 748, 775.
Lumière, 775.
Lying, 334, 435 f., 529, 741.

MacDougal, 775.
Mach, 214, 247, 554, 635, 698, 775.
Macht, 775.
MacLaurin, 775.
Magic of words, 185, 187, 202, 227, 260, 347, 408, 649.
Magnetic field experiment, 643.
Maier, 775.
Malinowski, 3, 19, 501 f., 775.
Manhood of Humanity, 7, 38 ff., 73, 91, 107, 181, 239, 264, 378, 381, 539, 727.
Maps, 11, 58, 247, 498, 750 f.

Marinesco, 542.
Marriage, infantilism in, 518.
Mast, 775.
Materialism, 521, 555 ff.
Maternal instinct, 127.
Mathematical abstractions, 67 ff.
'logic', 428.
methods, 130.
Mathematics, 204 ff., 568, 603.
as behaviour, 48, 71 f., 133, 218, 247, 289, 304, 565, 647, 752.
central principles of (table), 252.
defined, 250 ff.
evolution of, 257.
language of, 50, 66, 253 ff., 288, 293, 299, 401, 681, 752.
non-elementalistic character of, 288 ff.
Polish school of, 94, 541, 748.
schools of, 748 f.
structure of, 247 ff.
Matrices, 266 f., 705 ff., 714.
'Matter', 223 ff., 318, 383 ff., 573, 637, 648, 653, 679, 683 ff., 695, 698 ff., 719.
Maupertius, 589.
Mauthner, 775.
Maxima and minima, 588 ff.
Maxwell, 642, 644, 663, 673, 680, 686, 689, 695 ff.
Mayer, 541, 658, 681, 771, 775.
McClendon, 775.
McCollum, 126 f., 775.
McCracken, 125.
McCrea, 775.
McDougall, 775.
Meanings, 19 ff., 33, 74, 80 ff., 93, 137 ff., 142, 292 ff., 389 ff., 488.
Measurement, 251, 259, 292, 621, 648 ff.
Mechanics, 680 f., 694, 723, 730.
quantum. *See* Quantum.
Mechanisms of multiordinality. *See* Multiordinality.
'Mechanistic' vs. 'machinistic', 453 ff.
Medes, 775.
Medicine, 39 f., 178, 491, 531, 542.
Mellon, 49.
Memory, 291, 415 ff., 427, 456.
Menger, 775.
'Mental' activity, 119 f. *See also* Cortex and Thalamus.
diseases, illustrations, 128. *See also* 'Insane'.
hygiene. *See* Semantic hygiene.
Merz, 775.
Meshes, 619 ff.
Metabolism, 104, 115, 401, 630. *See also* Nutrition.
Methodology, 64, 70, 93 f., 130, 140, 142, 157, 176, 221, 242, 272 ff., 286, 303, 307, 310, 381, 425, 446, 451, 568, 592, 602, 611 ff., 632.
Meyer, 76, 303, 491, 537, 775.
Michelson-Morley experiment, 652, 654, 673.
Migraine, 128.
Mikami, 775, 778.
Militarism, 521, 558 f. *See also* Wars and World War.
'Mind and body'. *See* 'Body-mind'.
Minkowski, 57, 91, 99, 106, 223, 234, 243 f., 293, 318, 541, 575, 629 f., 640, 657, 665 ff., 672, 677, 718, 721, 726, 775.
Minima. *See* Maxima.
Mistakes, neurology of, 529 f.
Mnemiopsis, experiments with, 348.
m. o. *See* Multiordinal and Multiordinality.
Modality, 461, 759.
Molecule, 111, 322, 685.
Monakow, 775.
Money, 76, 519, 549.
Monkeys, experiment with, 349. *See also* Apes.
Moore, 368.
Morgan, de, 3, 367, 750, 775 f.
Morgan, C. L., 776.
Morgan, T. H., 769, 776, 778, 781.
Morons, 129, 291, 296, 309, 316, 485, 545, 676, 727.
Moving pictures, 230, 292, 578.
Mühl, 776.
Muller, 117, 776.
Multiordinal mechanisms, 74, 347, 439 ff., 455.
terms, 14, 22, 63, 74, 248, 253, 275, 389 ff., 416, 430 f., 433 ff., 451, 681, 711, 725, 753, 755.
Multiordinality, 14 f., 22, 63, 74, 94, 136, 145, 306, 324, 387, 432 ff., 449, 455, 482, 499, 725, 749 ff.
Multiplication, commutative and non-commutative, 710, 713 f.
Murphy, 776.
Muscles, antagonistic, 349.

N. *See* Newtonian and Trilogies.
$\bar{N}$. *See* Non-newtonian and Trilogies.
n-valued scheme, a diagrammatic summary, 463.
Names and substantives, 372.
Napier, 574.
Narcissistic period, 494, 496, 536. *See also* Sex disturbances and Infantilism.
Nationalisms, 518, 558, 561.
Natural order, 109, 169 ff., 219, 329, 446, 457, 470, 478 ff., 513, 755.
'law', 268, 286 f., 378, 385, 730.

Negative excitation. *See* Excitation.
premises, non-identity, 10, 60, 62, 139, 247, 268, 365, 396, 399 ff., 416 f., 427, 432, 441, 544, 545.
Nervous conflict, 529 f.
diseases (illustrations), 128.
function, 18, 115, 119, 365, 376 f., 511, 528, 630 ff., 637, 671, 702, 725, 758. *See also* Copying.
functions, order of, 170, 180 f., 297, 304, 330 ff., 479, 718.
impulses, velocity of, 156, 161, 192 f., 235, 422, 474, 756.
system, irritability of, 127, 349.
maturation of, 88, 175, 198, 320 ff., 327 ff., 510.
as a protective mechanism, 329, 499 f.
semantic education of, 149, 356.
structure of, 36, 50, 104, 130, 151, 156 ff., 163, 169, 177 ff., 182 ff., 192 f., 198, 224, 230, 235, 278, 287 ff., 292 f., 295 ff., 307, 320 ff., 327 ff., 340 ff., 360, 456, 529, 544, 553, 582, 604, 613 ff., 637, 647, 665 f., 671, 688, 716, 724 f.
Neurological function of abstracting, 235 ff., 278 f., 293, 325, 343, 354, 356, 363, 384, 389 ff., 413 ff., 438, 576.
problems, 88, 149, 156 f., 175 f., 193, 235 ff., 287, 320 ff., 456, 466, 475, 530, 553.
Neurology of mistakes, 529 f.
non-aristotelian, 157, 345.
Neurosis, 298, 334, 496, 502.
Newton, 27, 58, 85 f., 153, 162, 223, 244, 347, 574, 595, 626, 635, 641, 646, 671, 694, 730, 735 f.
Newtonian system, 27, 192, 232, 244, 347, 674, 679, 756. *See also* Trilogies.
Nicod, 776.
Non-additive, 37, 89, 116, 171 f., 188, 265, 313, 327, 354, 356, 528, 611 ff. *See also* Additivity.
Non-all, 375, 379, 417, 429, 469 ff., 475, 482, 545, 758. *See also* Allness.
Non-aristotelian biology. *See* Biology.
International Congresses, 555, 560.
Library, 554 ff.
neurology. *See* Neurology.
punctuation, 15, 16.
Review, 555.
Society, 554 ff., 560.
system, 7 ff., 27 f., 34, 42, 73, 91 ff., 107, 110, 131 ff., 139, 144 ff., 156, 184, 187 f., 198, 203 ff., 226, 273 ff., 281, 287, 316 f., 329 ff., 344, 365, 373, 379, 399, 403, 405, 411, 429 f., 437 ff., 452, 473, 486, 514, 530, 532 ff., 537 ff., 542, 574, 602, 663, 728, 747 ff. *See also* Trilogies.
Non-chrysippian 'logic', 92, 747 ff.
Non-el. See Non-elementalism.
Non-elementalism, 21, 32, 93, 96 f., 105 ff., 128 ff., 149, 155, 182, 188 ff., 192, 276, 285, 288 f., 304, 380, 394, 404, 420, 425, 429 ff., 499 f., 475, 512, 539 ff., 545, 568, 655, 699, 727 f., 748 ff., 757, 760. *See also* Elementalism.
Non-elementalistic character of mathematics, 288 ff.
Non-euclidean geometries, 27, 32, 64, 91, 233, 399, 604, 642 f., 658, 669, 701. *See also* Trilogies.
Non-identity, 50, 396, 417, 421, 427, 429, 432, 441 ff., 540, 545, 726.
Non-linear. *See* Non-additive.
Non-newtonian system, 27, 32, 91, 156, 203, 347, 379, 604, 643, 678 f., 701. *See also* Trilogies.
Nordmann, 776.
'Normal,' 41, 196, 335 ff., 341, 363, 460, 509, 515, 568.
Northrop, 776.
Number as a class of classes, 51, 208, 255.
definition of, 47, 208, 255 f., 258.
non-aristotelian definition of, 205 ff., 256.
evolution of, 48, 261, 438, 603.
imaginary, 354.
irrational, 247, 249, 259, 263, 266, 598.
negative, 354, 364.
notations for, 17, 256 f.
as operators, 355.
transfinite, 207.
Nunn, 776.
Nutrition experiments, 126 ff.

Objectification, 231 f., 317 f., 417, 443, 452 f., 458, 468, 650, 652 ff., 666, 675, 693, 695, 701.
Objects as dynamic processes, 382, 387.
Observer, ideal, 444 f., 447, 660.
Oedipus complex, 501.
Ogden, 33, 52, 542, 763, 776.
Onanism, 524, 526. *See also* Autoerotism and Sex disturbances.
Ontology, 521.
Operational methods, 2, 3.
Operators, numbers interpreted as, 267, 355, 588, 608, 713, 718.
Optics, 720 ff.
Order, 44, 57, 60, 93, 131, 151 ff., 155 f., 169, 179, 189 ff., 219 ff., 258, 316 f., 374, 380 ff., 428, 449 ff., 470, 474, 480, 545 f., 607, 639, 658, 665.
of abstractions. *See* Abstractions.
intension and extension, 171 ff.
and the nervous system, 59, 157 ff., 180 ff., 224, 330 ff.
language of, 664.

multi-dimensional, 30, 58, 96, 161 ff., 183 f., 192, 203, 248, 316 f., 319, 324, 349, 453, 459, 546, 639.
natural. *See* Natural.
neurological simplicity of, 59.
psychophysiological character of. *See* Psychophysiology.
recurrent, cyclic, 180 ff.
reversed, 44, 110, 164 f., 169 f., 180, 184 ff., 198, 329, 407, 428, 440, 450, 458, 480, 488, 752, 755.
Ordinal terms, 760.
Orent, 775 f.
Organism - as - a - whole, 9, 101, 103 f., 106 ff., 123 ff., 149, 156, 161 ff., 173, 182, 188, 192, 240, 288, 308 f., 376, 412 ff., 426 f., 456, 469, 475, 498 ff., 523, 531 f., 545 f., 552, 606, 638, 647, 727.
Orthological institute, 52, 763.
Osborn, 776.
Osgood, 566, 776.
Oswald, 112.
Over-stimulation, protection from, 169, 500.
Ownership, 549.

Pacifism, 559.
Packard, 776.
Paget, 776.
Painter, 776.
Panics and misevaluations, 49, 338.
Paradox, Tristram Shandy, 210.
Paradoxes, 738 ff.
Paranoia, 87, 174, 200.
Pareto, 776.
Parker, 776.
Particle orientation, 700.
Participation, law of, 514.
Pascal, 567.
Paterson, 511, 774, 776.
Paton, 776.
Pauli, 111.
Pavlov, 18, 152, 183, 195, 315 ff., 326 ff., 332 ff., 341, 356 ff., 369, 452, 459, 469, 488, 512, 767, 776.
Peano, 131, 154, 251, 433, 748, 776.
Pearl, 776.
Pearson, 776.
Pedophilia, 524. *See also* Sex disturbances.
Peirce, 4, 777.
Pellagra, 126 f.
Perception, 66, 253, 457.
Periodic functions, 230, 319, 597 ff., 629.
Personalities, multiple, 149.
Peterson, 770, 777.
'Philosophy', 77 ff., 87, 90, 139, 172 f., 176, 180, 215, 227, 231, 241, 244, 293, 302, 310, 371, 453, 521, 563, 606, 695.
Phobias, 337, 368, 574.
Physical subjectivity, 244.
Physics of high pressure, 249, 541 f.
Physiological functions, 12, 44, 264, 274, 304, 316, 321 f., 424, 474, 491.
gradients. *See* Dynamic gradients.
Piaget, 777.
Piaggio, 777.
Picard, 133.
Piéron, 1, 2, 4, 19, 99, 131, 151, 188, 268, 313, 369, 386, 452, 603, 613, 615, 777.
Pierpont, 777.
Piezochemistry, 541 f.
Pitkin, 777.
Pituitary gland, hyper- and hypo-pituitarism, 129.
Pity, 440.
Planaria torva, experiments with, 123.
Planck, 2, 563, 688, 700, 705, 714, 720 f., 777.
Planck's constant, 714 ff., 721.
Plato, 87 ff., 203, 726.
Plauson, 777.
Plenum, 102, 229 ff., 319 ff., 345, 480, 576, 622, 625, 666, 700, 718.
'Plus' feeling, 605 ff. *See also* Additivity.
Poincaré, 1, 2, 7, 76, 79, 85, 99, 107, 188, 204, 223, 367, 615, 747, 767, 777.
Point-event, 219, 383, 482.
Polakov, 777.
Politicians, 559.
Polygordius, experiments with, 124.
Polysynthetic languages, 463.
Popularization of science, 55 f., 484, 555.
Postulate systems, 144, 275, 277.
'Practical' man, 516, 549.
Pragmatism, 110.
Prantl, 777.
Preference, problems of, 442.
Preferential paths (Bahnung), 235, 299.
Pre-logical 'thinking', 514.
Premises, 10, 28, 50, 60 ff., 254, 308 f.
Prevention, 175, 187, 439 f., 443, 446, 451, 487, 490, 493, 501.
Prince, 128, 777.
Principia Mathematica, 56, 137, 409, 430 ff., 565, 737 ff.
Probability, 92 ff., 210, 215 f., 310, 405, 461, 602, 680, 716 f., 724.
Processes, 96, 139, 205 ff., 218, 227, 350, 352, 362, 458, 464, 495.
of abstracting. *See* Abstracting.
nervous, neural, 291, 297, 307, 317, 360. *See also* Nervous.
and order, 191, 316, 467.
periodic and rhythmic, 230, 319.
semantic, 229, 281 f., 284 f., 302, 378, 401, 507.
and structure, 162.
survival, 294, 402.

unit, 211 f.
world-event, dynamic, 263, 382, 387.
Proclus, 504.
Projection, mechanism of, 80 ff., 90, 170, 176, 317, 463 ff.
Propaganda, 446.
Proportionality, 586, 603, 609 f.
Propositional function, 94, 136 ff., 149, 275, 277, 286, 436 f., 738 ff.
Propositions, 93 f., 430 ff., 738 ff., 745 f.
Prostitution, 269, 305, 524 f.
Protection from over-stimulation, 169, 500.
Protoplasm, 102 f., 114 ff., 120, 165 f., 235, 321, 323, 342, 345, 380, 493.
Protothetic, 748.
Psychiatry, 30, 121, 128, 130, 138 f., 149, 163, 181, 185, 187 f., 227, 278, 288, 298, 303 f., 337, 342, 348, 358 ff., 378, 439 f., 450, 490, 493, 500, 503, 512, 521, 523, 541 ff., 568.
Psychobiology, 303.
Psychogalvanometer, 119, 196, 335, 508.
'Psychology', 70 ff., 95, 192, 204, 244, 256, 281, 289, 303 f., 336, 344, 401, 462, 559, 699, 726, 749, 752.
gestalt, 74.
Psycho-logics, 23 f., 31, 35, 72, 95, 133, 149, 219, 270 ff., 287 ff., 296, 310, 342, 378, 440, 490, 539, 582.
Psycho-neural means, 296, 304, 306 f., 425, 431.
Psychoneuroses, 129, 298, 334, 498, 502 ff. *See also* Neuroses.
Psychophysiology, 17 f., 23 ff., 93, 250, 281, 306, 313 ff., 425, 449, 451, 503, 515, 536, 540.
Psychophysiological character of order, 319, 321 f., 446, 449, 451, 545.
Psychoses, 298, 496, 498, 511.
Psychotherapy, 24, 34, 88, 104, 119, 147, 185, 197, 298, 358 f., 364, 441, 443, 450, 467, 500 ff., 533 f., 536.
Ptolemy, 504.
Punctuation, non-aristotelian, 15, 16.
Pythagorean theorem, 615, 622 ff., 646, 651, 668.

Quality, 250, 264.
Quanta, 2, 134, 249, 319, 322, 386, 639, 686, 705.
Quantity, 708.
Quantum mechanics, 30, 210, 214 f., 242, 249, 326, 354, 453, 487, 541, 550.
theories, 2, 7, 60, 91, 135, 192, 210, 214 f., 241 f., 263, 265, 267, 322, 354, 381, 386 f., 603 f., 630, 680, 683, 687 ff., 695, 698 ff., 711, 714 ff., 720, 725, 760.
Question-begging epithets, 403, 446, 496.

Rabaud, 777.
Radiant energy from cells, 120.
Radio-activity, 686, 689.
Rainich, 282, 384, 569, 571, 665, 777.
Ramsey, 777.
Rank, 771, 777.
Rashevsky, 777.
Rationalization, 368, 448, 458, 465 f., 500 f., 524, 526, 604.
Rats, 127, 289 f., 303, 350.
Rays, 117 f., 236 f., 244, 686 f., 689.
Recall, 361.
Recognition, 507.
Reducibility, axiom of, 433.
Reference, frames of, 242, 616 ff., 729, 731.
Reflex mechanisms, 158 ff., 481.
Reflexes, 19, 151, 315 ff., 326 ff., 341, 353, 360, 455 ff., 488, 512.
Regeneration, 124.
Regression, 193, 200, 296, 305, 321, 332 f., 337, 378, 495 ff., 501, 503, 508, 515.
schematic representation, 497.
Reiser, 777.
Relating, process of, and associations, 284, 289, 512, 613.
Relations, 57, 60, 69, 93, 103, 131, 155, 161 ff., 173, 188 ff., 220 ff., 261, 287, 453, 510, 545, 625.
asymmetrical, 56, 188 ff., 197 f., 202 f., 220, 258, 317, 374, 406 f., 438, 452, 480, 640, 658, 711.
similarity of, 57.
structure, order and, 161 ff., 253, 260 ff., 269, 284 f., 299.
symmetrical, 189, 191, 194, 258, 438.
Relativity, theory of, 2, 97, 234, 386, 408, 603, 667, 729 ff., 734.
old mechanical, 652.
Repression, 343, 358, 361, 421, 481, 529.
Reproductive glands, 522.
Rhythmic processes and tendencies, 114, 123, 230, 348.
Richards, 33, 542, 743, 776 f.
Richter, 511, 768, 774, 777.
Rickets, 118, 126.
Riemann, 86, 265, 286, 573, 622, 642 f., 646, 739.
Rietz, 777.
Rinaldo, 777.
Ritchie, 777.
Ritter, 101, 777.
Roback, 777.
Robb, 777.
Roberval, 574.
Römer, 644.
Röntgen rays, 117, 236 f., 244, 686.
Roots, radiations from, 120.
'learning' of, 120.
Rotating fan, experiment with, 382 f., 472, 694.

Royce, 58, 131, 220 f., 751, 778.
Ruark, 778.
Rueff, 778.
Rulers, 41, 77, 269, 295, 549, 559.
Runge, 765.
Russell, 53, 56 f., 66, 85, 94, 131, 136 f., 151 f., 154, 188, 210, 214, 216, 219, 223, 234, 251, 255, 311, 428 ff., 434, 491, 565, 635, 737 f., 748, 753, 778, 780.
Russian revolution, (1905), 521.
Rutherford, 685, 687 ff.
Rutzler, 768, 778.

Saccheri, 505.
Sanity, 46, 84, 114, 130, 150, 161, 183, 192, 200 ff., 225, 232, 288, 297 f., 306, 309, 311, 316 f., 328, 355, 371, 429, 434, 438, 440, 443, 462, 474, 480, 484, 491 ff., 508, 531 f., 545, 547, 574 ff., 632, 647, 655, 693, 701, 727, 755, 761.
  theory of, 23, 46, 75, 149, 183, 287, 293, 316 f., 377, 412 ff., 434, 440, 469, 524, 545, 712.
Santayana, 1, 3, 4, 253, 537, 778.
Schiller, 778.
Schizophrenia, 87, 175, 185, 368, 511, 516, 518, 524, 568. *See also* Language of.
Schlick, 778.
Schrödinger, 247, 603, 685, 698, 717 f., 722 ff., 778.
Science, popularization of, 55 f., 484, 555.
Scientific era, 18, 194.
Scott, 36.
Scurvy, 126 f.
Sea urchin, experiments with, 124.
Selection and abstracting, 507.
Selfishness, Anglo-Saxon philosophy of, 521.
Self-reflexiveness, 58, 323, 739, 751.
Semantic aphasia, 19 f.
  blockage. *See* Blockage.
  compensation, 201.
  considerations, 'matter', 'space', etc., 241 ff.
  defined, 19.
  disturbances, 62, 82, 84, 87, 109, 138 f., 183, 192, 198 ff., 202, 219, 224 f., 227, 231 f., 260 f., 265, 269, 278 f., 288, 294 f., 297, 302, 304, 306, 311, 320 f., 332 f., 336, 382, 385, 407, 417 ff., 421, 434, 443, 456, 460, 478, 488 f., 496 ff., 510, 528, 531, 546, 567 f., 573, 638, 650, 654, 666, 671 f., 674, 676 f., 681, 697, 716, 727, 757, 761.
  epidemics, 200.
  evolution, 194.
  flight from 'reality', 199.
  flexibility, 207, 338 ff., 443, 451, 461, 547.
  hygiene, 17, 31, 175, 287, 489, 530, 560.
  periods of growth, 494 ff.
  problems, international, 270 ff., 525 f.
  reactions, 19 ff., 93, 97, 115 f., 121, 135, 138, 163, 176, 234, 269, 291, 336, 338, 356, 364, 378, 389 ff., 491, 501, 566.
  description of, 24.
  training of, 18 f., 27, 32, 287.
Semantics, general 8, 95, 250, 265, 380, 434, 461 f., 474, 535, 540, 545, 752.
Sencourt, 778.
'Sensations', 235 ff., 278 f., 494, 513.
'Sense' organs, 235 ff., 278 f., 459.
Sex disturbances, 305, 524. *See also* Autoerotism, Exhibitionism, Homosexualism, Infantilism, Onanism, Pedophilia.
  glands. *See* Gonads.
Sharks, experiments with, 346.
Shaw, 251 f., 267, 283 ff., 778.
Sheffer, 94, 145, 778.
Sheppard, 778.
Sherrington, 778.
Sierpinski, 778.
Signals and symbols, 333, 334, 338.
Significs, 33.
Silberer, 778.
Silberstein, 778.
Silence on objective levels, 35, 399, 404, 410, 416, 446, 468, 477, 481, 485, 637, 644, 656. *See also* Un-speakable levels.
Silk worm moths, experiments with, 125.
Similarity vs. difference, 166 f., 381, 514.
  of structure. *See* Structure.
Simultaneity, 2, 649, 659 ff., 670, 732 ff.
Skarzeński, 748, 753.
Smith, D. E., 775, 778.
Smith, H. B., 781.
Smith, W. B., 778.
Socialism, 270 ff., 486.
Sommerfeld, 569, 635, 692 ff., 705, 720, 778.
Sommerville, 778.
'Space', 57, 99, 106, 154, 184, 191 f., 218, 222 ff., 289, 299, 316, 318, 383 f., 573, 575, 591, 637, 647 f., 652 f., 665 ff., 677, 693, 695, 701, 715, 719.
Space-time, 57, 99, 106, 108, 130, 184, 191, 289, 316 ff., 383 ff., 407, 513, 541, 626 ff., 667 ff., 679, 715 f., 721, 724, 734 ff.
Space-time, curvature of, 233, 243, 318, 384, 386.
Spalding, 36.
Spaulding, 778.
Spearman, 778.
Spell-marks, 138, 142.
Spengler, 47 ff., 778.
Spiral theory, 181.

Splitting, 27, 30, 65, 149, 181, 184, 231, 484, 498.
Sponges, experiments with, 346.
s.r. *See* Semantic reactions.
St. Paul, 753.
Starfish, experiments with, 347.
Starling, 778.
Static vs. dynamic, 139 f., 224, 240, 297 ff., 306, 324, 454, 647.
Statistical laws, 716.
  macroscopic averages, 218, 680, 724.
  mass effect, 376.
Statistics, 310, 376, 716, 724.
Stekel, 778.
Sterility in females, 127.
Stewart, 778.
Stieglitz, 778.
Stiles, 778.
Stimuli, never simple, 317, 319.
  and inhibitions. *See* Inhibitions.
Storch, 778.
Stratification, vertical and horizontal, 130, 395 f., 410, 441, 468 f., 474, 476, 503 f., 507, 545, 693.
Structural calculus, 259.
Structural Differential, 13, 26, 329, 386 ff., 397, 399, 403, 410 ff., 416 ff., 423, 426 ff., 443 ff., 453 ff., 469 ff., 481, 485, 487, 501, 574, 758.
  unconscious, 361, 506, 535.
Structure, 55 ff., 62 ff., 88, 93, 96, 101, 104, 109 f., 113 ff., 121, 130, 140 ff., 155, 161 ff., 191, 194, 197, 200 f., 217 f., 221 f., 224 ff., 232, 234 f., 247, 268 f., 275, 277, 280, 284 f., 287, 304, 306 ff., 317 ff., 340, 348, 351, 407, 452, 468, 483, 530, 545, 551, 575, 592, 664 ff., 701, 751.
  aristotelian, 59, 92, 141, 302.
  as the only content of knowledge, 20, 22, 194, 224, 260, 324, 326, 348, 544, 566, 640, 697, 699, 725, 751.
  defined, 56.
  of events, 530.
  and function, 135, 322 f., 332 f., 544 f.
  of knowledge, 33, 220 f., 223, 239, 269, 433, 438 ff.
  language of similar, 42, 47, 58, 64, 241, 292, 501.
  linguistic, 30, 42 ff., 55, 59 ff., 70, 107 ff., 129, 144, 191, 199 f., 222, 241, 243, 253, 268 ff., 273, 275, 279, 295, 318, 324, 337, 344, 355, 363, 372 f., 379, 383 ff., 400 ff., 425, 433, 464, 468, 473, 485, 505, 513 f., 544, 554, 613, 618, 632, 638, 650, 653, 655 ff., 675, 677, 681 695, 725, 749 ff.
  as a link between the objective and verbal worlds, 20, 59, 61, 365, 751.
  of mathematics, 43, 69, 275, 306 ff., 353.
  and multi-dimensional order, 161 ff., 316, 685.
  linguistic, and postulate systems, 144.
  similarity of, 57 f., 60 ff., 164, 224, 262, 282 f., 287 ff., 293, 295, 322 f., 328 ff., 385, 412 ff., 441, 540, 566 f., 699.
Strychnine, effect of, 349.
Stupor, emotional, 217, 676, 681.
Sturtevant, 776, 778.
Subject-predicate, 62, 85, 92, 131, 189, 190, 224, 306, 371.
Subjectivity, physical, 244.
Sublimation, 181 f., 304, 439 f.
'Substance' and function, 163, 543, 565.
Suggestions, non-aristotelian, 555 ff.
Sullivan, H. S., 778 f.
Sullivan, J. W. N., 779.
Surfaces, theory of, 242, 260, 568, 619, 621 ff., 642, 667.
Survival, 35, 44, 168 f., 176 f., 179 f., 184, 187, 193, 196, 198, 239, 278, 293 f., 296 f., 305, 328, 336, 342, 344, 402, 459, 480, 579, 727, 755.
Sylvester, 264.
Symbols, 76 ff., 84, 138 ff., 202, 207, 296, 333 f., 338, 387 ff., 475, 519, 549. *See also* Correct symbolism.
Symptoms of illness, 118 f.
Synapses, described, 157 ff.
System-function, 145, 148 ff., 533 f.
System-structure, 146 f.
Syz, 773, 779.

Talmud, i.
Tarski, 93, 210, 281, 310, 461, 748, 779.
Tau effect, 324.
Teleology, 521.
Tennessee trial, 40, 166, 279, 302.
Tensors, 69, 146, 242, 260, 268, 540 f., 565, 568, 571, 617 ff., 627 ff., 656.
Terms, undefined, 21, 93, 152 ff., 155, 301, 310, 344, 418 ff., 505.
Tetanus bacillus, effects of, 349.
Thalamic 'thinking', 290 ff.
Thalamus, 157, 188, 193, 268, 290 f., 299, 369, 422, 456, 460, 505, 511, 758.
Thalbitzer, 779.
Theism, 521.
Theories, 110, 277 ff.
Theory of the absolute, 640.
  of aggregates, 69, 208, 280 ff., 419, 428.
  of Einstein. *See* Einstein.
  of Freud, 491 ff., 501.
  of function, 266, 275, 281, 575, 726.
  of groups, 69, 266 ff., 280 ff., 607, 658.
  Hilbert proof, 204.
  of implication, 182.

of invariance, 267 f., 287.
of knowledge, 554. *See* Epistemology.
of matrices, 267. *See* Matrices.
of numbers, 267.
of operators, 267, 283 f. *See* Operators.
of probability, 210, 310. *See* Probability.
physical, 282.
quantum. *See* Quantum.
of sanity. *See* Sanity.
of statistics, 310.
of surfaces. *See* Surfaces.
of types, 429 ff., 737 ff., 753, 759. *See also* Types.
unified field, 378, 386, 541.
Thomas, 779.
Thompson, D. W., 779.
Thompson, S. P., 571, 602, 779.
Thomson, G. P., 781.
Thomson, J. J., 686, 689.
Thorndike, E. L., 779.
Thorndike, E. M., 773.
'Thought', 99. *See also* Thalamic 'thinking'.
'laws' of, 197, 210, 310, 404, 747 ff., 760.
Thurstone, 779.
Thymus, 522 f. *See also* Endocrine glands.
Thymopaths, 523.
Thyroid, 128 f., 522 f. *See also* Endocrine glands.
Thysanozoon Brochii, experiments with, 123.
Tilney, 779.
'Time', 57, 99, 106, 154, 184, 191 f., 218, 222 ff., 289, 299, 316, 354, 375, 383, 537.
Time-binding, 8, 39, 108, 183, 231, 239, 279, 285, 292, 294, 298, 318, 321, 354, 376, 378, 381, 394, 503, 526, 539, 545, 548, 557 f., 573, 591, 601, 626 ff., 637, 645, 647 f., 652 f., 666 ff., 677, 693, 695, 701 f., 715, 719.
mechanism of, 40, 52, 285, 291, 503, 539, 604, 638, 699.
Tolerance, 518.
Tolman, 760, 779, 781.
Tortoise, decerebrated, 511.
Totalities, illegitimate, 430, 753.
Tramer, 779.
Transformation, 131, 266, 268, 281, 284, 286, 616 ff.
Lorentz-Einstein. *See* Lorentz-Einstein.
Translation of dynamic into static and vice versa, 224, 240, 288, 293, 297 ff., 300, 304, 306 ff., 324, 454, 573, 582, 632, 638, 647, 671, 758.
of higher to lower orders and vice versa, 240, 304, 454, 460, 482, 546, 632, 638, 647.
Trial and error, 272, 274, 305.
Trigonometry, 597 ff.
Trilogies, $A\ E\ N$ and $\bar{A}\ \bar{E}\ \bar{N}$, 7, 27, 91, 94, 97, 142, 233, 241, 379, 506, 527, 606, 675, 749, 756.
Tristram Shandy paradox, 210.
Tropism, 101 ff., 108, 327, 541.
chamber, 120.
Tschuppik, 779.
Turnbull, 779.
Tyler, 779.
Types, mathematical, 311, 428 ff., 432, 737 ff., 753, 759.
of order, 151.
theory of, 429 ff., 737 ff., 753, 759.
Uhlenbeck, 692 ff.
Ultra-violet rays, 117 ff., 236 f.
Uncertainty, general principle of, 93, 310, 405, 540, 760.
restricted principle of, 107, 310, 541.
Unconscious, 491 ff.
assumptions, 504.
structural, 361, 506, 535.
'Unknowable', 754.
'Unicorn', the, 81, 137.
Unified field theory, 378, 386, 541, 667.
United States of America, 38, 270 ff., 302, 522, 526.
United Socialistic Soviet Republics, 270 ff.
Universal agreement. *See* Agreement.
languages, 52, 763.
laws, 287, 619, 626, 640, 689.
Universe of discourse, 225 f., 254.
Universe, the and a, 86.
Un-sanity, 41, 105, 279 f., 294 ff., 299, 304, 306, 329, 397, 484, 491 ff., 574.
Un-speakable levels, 34 f., 62, 202, 226, 278, 306, 317, 323, 328, 372 f., 405, 426, 432, 444, 447, 476 ff., 480, 484, 544, 566, 637, 644, 656, 694, 751 ff.
Urey, 778 f.
Vaihinger, 779.
Values, delusional, 35, 40, 195, 198, 228, 408.
general theory of, 453, 554, 759.
Van Woerkom, 131.
Variables, 93, 134 ff., 207, 286, 575, 585, 601, 631, 671, 706, 717, 756, 759, 761.
Variation, Hamilton's principle of, 722.
Vasiliev, 779.
Veblen, 151, 245, 615, 779.
Vectors, 146, 354, 495, 609 ff., 617 ff., 710, 713.
Velocity, 102, 156, 162, 192, 213, 235, 263, 591 f., 627, 649, 654, 658, 661, 668, 677, 703, 712, 721, 733. *See also* Light.

of propagation, 641 ff., 648, 651, 656, 660, 673, 677, 756.
Vendryès, 779.
Vibrations, table of, 237.
harmonic, 601, 629, 704.
Vicious circle fallacies, 270, 295, 430 f., 737 ff.
Vulliamy, 779.
Visual types, 460.
Visualization, 229, 278, 300, 304, 307, 386, 423, 446, 452 ff., 458 ff., 664 f., 715, 718, 724.
Vitamins, 118, 126 f.
Voigt, 676.
Voting, 560.

Wagner, 779.
Wallis, 505.
Wars, 727. *See also* Militarism and World War.
Washburn, 779.
Watson, G. A., 320.
Watson, J. B., 527, 779.
Wave-length, 723.
Wave theory, 108, 249, 597 ff., 647, 700, 717 f., 722, 725.
Waves, advancing plane, 601.
electromagnetic, 644.
hertzian, 236.
table of vibrations, 237.
Weatherburn, 779.
Weber, 238, 644.
Weierstrass, 215, 218, 299.
Weinstein, 779.
Weiss, 737 ff., 780.
Welby, 33.
Weyl, 1, 64, 79, 85, 94, 204, 233, 249, 282, 573, 603, 609, 615, 635, 642, 739, 742, 747 f., 775, 780.
Wheeler, 3, 38, 66, 339, 780.
White, A. D., 780.
White, W. A., 76, 104, 124, 128, 185, 773, 780.
Whitehead, i, 2, 3, 7, 66, 76, 85, 137, 154, 219, 247, 251, 257, 367 ff., 371, 386, 390, 412, 426, 429, 565, 575, 675, 737, 741, 748, 780 f.
Whittaker, 567.
Wiener, 714.
Wilder, 780.
Williams, F. E., 780.
Williams, H. B., 245, 780.
Wilson, E. B., 772, 774, 780.
Wilson photographs, 689.
Windelband, 781.
Wittgenstein, 53, 56, 368, 431, 781.
Woodworth, 781.
Words, magic of, 185, 187, 202, 227, 260, 347, 408, 649.
World-line, 627 f., 666, 670, 734.
point, 666, 672.
of Minkowski, 665 ff., 679, 718, 726.
World War, 38, 44, 46, 269, 334, 446, 521, 549.
Worms, experiments with, 116, 123 ff., 346.
Worry, 440.
Wundt, 781.
Wüst, 771, 781.

X-rays. *See* Röntgen-rays.

Yealland, 781.
Yerkes, A. W., 781.
Yerkes, R. M., 124, 774, 781.
Young, J. W., 781.
Young, J. W. A., 781.

Zaremba, 781.
Zavadski, 358, 360 f.
Zeno, 71.

SCIENCE AND SANITY

AN INTRODUCTION TO NON-ARISTOTELIAN SYSTEMS AND GENERAL SEMANTICS

BY

ALFRED KORZYBSKI

(Author of *Manhood of Humanity*)

SCIENTIFIC OPINIONS ABOUT THE FIRST EDITION, 1933:

1. ANTHROPOLOGY ...	Professor B. Malinowski, University of London.
2. BIOLOGY	Doctor C. B. Bridges, Carnegie Institution. Professor C. M. Child, University of Chicago. Professor H. S. Jennings, Johns Hopkins University. Professor R. Pearl, Johns Hopkins University.
3. BOTANY	Doctor D. G. Fairchild, United States Department of Agriculture.
4. CONDITIONAL REFLEXES	Doctor W. H. Gantt, Phipps Psychiatric Institute, Johns Hopkins Hospital.
5. EDUCATION	Doctor E. L. Hardy, President States Teachers' College, San Diego, California. C. L. Williams, President Williams College, Berkeley, California.
6. ENTOMOLOGY	Professor W. M. Wheeler, Harvard University.
7. GENETICS	See C. B. Bridges, D. G. Fairchild, H. S. Jennings.
8. OPHTHALMOLOGY ..	Professor W. H. Wilmer, Johns Hopkins University.
9. MATHEMATICS	Professor E. T. Bell, California Institute of Technology.
10. MATHEMATICAL FOUNDATIONS AND LOGIC.	Bertrand Russell, F.R.S., London, England.
11. MATHEMATICAL PHYSICS	Professor B. F. Dostal, University of Florida.
12. NEUROLOGY	Professor C. J. Herrick, University of Chicago.
13. PHYSICS	Professor P. W. Bridgman, Harvard University. Professor R. J. Kennedy, University of Washington, Seattle, Washington.
14. PHYSIOLOGY	Professor R. S. Lillie, University of Chicago. Professor H. B. Williams, Columbia University.
15. PSYCHIATRY	Doctor P. S. Graven, Washington, D. C. Doctor J. A. P. Millet, New York City. Doctor M. Tramer, University of Bern, Switzerland. Doctor W. A. White, Superintendent Saint Elizabeth's Hospital, Washington, D. C.
16. SEMANTICS	See B. Malinowski.

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1. ANTHROPOLOGY.

BRONISLAW MALINOWSKI, Ph.D. (Cracow), D.Sc. (London), Professor of Social Anthropology, School of Economics, University of London.

"The functional or relational conception of matter, mind and, finally, of human culture, seems to be gradually crystallising from all attempts at scientific synthesis. Count Korzybski's work contributes to these efforts in no mean measure. I am perhaps biassed as a countryman, but to me this Polish attempt at synthesis seems to rank as one of the most important. I am of course unable to express a competent judgment on its mathematical, scientific—in the narrow sense of the word—and philosophical side. As regards however semantics and the anthropological issues discussed by Count Korzybski, I am in complete agreement with his approach. I should like to add that the approach is so new and fundamental that it will take some time for us to become completely familiar with it. For the present I should like to say that I have not yet mastered all the intricacies of Count Korzybski's system, so my appreciation must naturally be regarded as preliminary."

2. BIOLOGY.

DOCTOR CALVIN B. BRIDGES, Biologist, internationally known specialist in heredity, Carnegie Institution of Washington, in residence at California Institute of Technology, Pasadena.

"In several fields of Biological sciences the unconscious drift of thought has been for some years more and more in the direction which Count Korzybski designates as 'non-elementalistic.' Thus, the distinction, once thought fundamental, between heredity and environment, loses its force, and the organism is now redefined as the focus of both internal (hereditary) and external (environmental) activities. What is considered external rather than internal, changes with the point of view and the size of the unit (nations, man, gland, cell, nucleus, chromosome, gene) which is made the basis of the formulation. The formulation becomes relational, non-elementalistic, organism-as-a-whole. The reformulation of biological concepts is made ultimately inevitable, and is greatly hastened and aided in transition by the generalized point of view established in the non-aristotelian system of Korzybski. The elimination of 'identity' constitutes the first and most fundamental general step for such a non-aristotelian and non-elementalistic reconstruction. Following this complete reformulation and its application in science and in life, the psycho-logical and environmental conditions for man would be improved to such an extent that it is not yet possible to foresee the entire result in the new enviro-genetic manifold."

C. M. CHILD, Professor of Zoology, University of Chicago.

"I think that Count Korzybski has a view point of great interest and that his method of attack on the various problems with which he deals cannot fail to be of value."

H. S. JENNINGS, Henry Walters Professor of Zoology and Director of the Zoological Laboratory, Johns Hopkins University.

"The attempt of Count Korzybski to formulate the world and its processes, keeping in view as a guiding principle the fact that no two things are identical, seems to me of the greatest interest and value. It is something that had to be done, and it has within it the seeds of a much needed intellectual revolution."

RAYMOND PEARL, Professor of Biology, Johns Hopkins University.

"I have known and followed Count Korzybski's work for many years with the keenest interest. In this new book he makes, in my opinion, a contribution to human thought and understanding of the very first rank of importance. It states and develops a really new idea. The consequences of that idea will, in the passage of time, be far-reaching and fundamental. At long last real hope is offered of measurably freeing man from some of the dreadful consequences of his verbalistic bonds."

See also W. M. WHEELER.

3. BOTANY.

DOCTOR DAVID G. FAIRCHILD, Botanist, U. S. Department of Agriculture, Washington, D. C., plant explorer and pathologist.

"I am much impressed with the profundity of Korzybski's *Science and Sanity*. I find it hard to get out of the meshes of the old aristotelianism and wish that I had been able to read this book in my youth, for then I could have acquired the new language of relations.

Korzybski's masterly treatise will act as a powerful force in natural selection perhaps, when it brings into common use the non-aristotelian methods, for it will favor in most pursuits those who are capable of conceptual thought and confuse and eliminate those who want quick off hand decisions such as are usually blurted out with great show of confidence. Of one thing we may be assured: once a man grasps the general idea Korzybski is driving it, he cannot fail to look at the world of everyday language from a different standpoint.

Korzybski's criticisms are so profound that they change the very foundations upon which we have been used to depend. As I look back to my years of travel over the world I can see that I did everything I ought not to have done in the way of bad thinking (we all have done this I suppose; bad thinking must be the common plague of mankind).

These last years among West Africans in Africa and the West Indies have made me realize keenly that primitive man identifies about like the animals do and has no consciousness that he abstracts at all. Of course if we are not conscious of abstracting; in other words, if we copy primitives or animals in our nervous reactions, then I suppose any kind of maladjustment can be expected. The simple and efficient neuro-psychological non-aristotelian technique which Korzybski formulates in his Structural Differential for the purpose of the elimination of identification holds promises that we may finally outgrow the infantile stage of our civilization. I wonder that Educators have not already taken up this pressing problem and made the elimination of identification and the acquiring of consciousness of abstracting, the main aims of all education.

These impressions regarding Korzybski's remarkable book come at the close of many years of travel through the world,—in savage countries, in the Orient, in South America and South Africa,—and had they only been a part of my mental training before these travels, the results of my observations could hardly have failed to have been much nearer to the actualities.

I predict a steady conversion to the point of view of this most interesting and important work."

4. CONDITIONAL REFLEXES.

DOCTOR W. HORSLEY GANTT, Phipps Psychiatric Institute, The Johns Hopkins Hospital. Formerly for five years co-worker with Professor Pavlov in Leningrad.

"I have read with great interest Count Korzybski's *Science and Sanity* and feel that it is very important for science as well as general education and progress of human thinking. It expresses a point of view and a truth that I have not seen stated previously. I was particularly interested in the chapters dealing with conditional reflexes. Korzybski dicsusses the matter with profound and accurate understanding, and the suggestions he makes are most timely and helpful to those who are working in this field. Anyone interested in the broader aspects of science I am sure will find in Korzybski's book an original and far-sighted view of the whole modern teaching of the subject."

5. EDUCATION.

DOCTOR EDWARD L. HARDY, President State Teachers College, San Diego, California.

"Count Korzybski's *Science and Sanity* should be read by all persons seriously interested in or concerned with the next necessary steps in the development of educational principles and procedures."

CORA L. WILLIAMS, Mathematician, President Williams Institute, Berkeley, California.

"What Einstein has done for the outer realm of our being, Korzybski is doing for our inner realm. It is to be hoped that some understanding person will endow a chair of Non-aristotelian General Semantics for him in the Institute of Advanced Study so that these two lines of research may go on together."

6. ENTOMOLOGY.

WILLIAM MORTON WHEELER, Professor of Entomology, Harvard University.

"Count Korzybski's work seems to me to be of great interest and value not only to the lay reader but also to the student of science and to the biologist and sociologist in particular for three reasons. First, his views give greater generality to the significance of the organism as a whole, and of structure and creative synthesis, or emergence, which are being increasingly emphasized by biologists, psychologists and sociologists working in the most diverse fields. Second, the sections of his work dealing with the intellectual vices of wishful thinking, verbalism and identification, to which we are all more or less addicted, point the way to the acquirement of mental balance and sanity. And third, his method of attaining this sanity through a non-aristotelian system and a realization of the meaning of the abstractions and symbols which we are constantly using, lays the foundation for a sound and much-needed social, commercial and political ethics."

7. GENETICS.

See CALVIN B. BRIDGES, D. G. FAIRCHILD and H. S. JENNINGS.

8. OPHTHALMOLOGY.

WILLIAM H. WILMER, M.D., Professor of Ophthalmology, Johns Hopkins University and Ophthalmologist-in-chief, Johns Hopkins Hospital.

"Count Korzybski's viewpoint is very unique, fascinating, and, I think, very logical. The induction of non-identity would cover a great many ills, mental, moral and physical. For more than a quarter of a century, I have observed the retrogression of a number of great men after a certain time; and I feel that their failure to hold their greatness has been due largely to an egocentricism. What is true of these men whom the world called great for awhile, is equally true of the masses of humanity, who have not attained to greatness. Many of these could probably have been saved by the proper attention to psychophysiology."

9. MATHEMATICS.

E. T. BELL, Professor of Mathematics, California Institute of Technology.

"I think it is obvious that Korzybski is working in a direction of the highest present importance for science and life. This is the more so as some sort of corrective seems to be needed for the well-meaning but ill-considered popular announcements by certain leading scientific men.

A little careful consideration of the recognized fundamentals of scientific and other thinking, such as Korzybski's book aims to set forth clearly, would prevent such really futile pronouncements by prophets of science and make the public more chary in swallowing every transient guess.

Korzybski, among personal contributions of his own concerning the law of identity, has succeeded incidentally in making current the fundamental revolution in mathematical and other basic thinking, which goes under the name of a non-aristotelian logic, and bringing to educated people an account of the most significant advance in abstract thought of the past millennium. The profound modifications of rational, mathematical thinking which began about thirty years ago with the work of Brouwer, have, so far as I am aware, escaped the notice of those who undertake

to report science and mathematics to the general public. The reader of Korzybski's book will gain an outlook on these new fields as well as an insight into the author's contributions to the problem of identity. Brouwer challenged one of the laws of Aristotle, Korzybski challenges another."

See also P. W. BRIDGMAN, B. F. DOSTAL, R. J. KENNEDY, BERTRAND RUSSELL, M. TRAMER, C. L. WILLIAMS, H. B. WILLIAMS.

10. MATHEMATICAL FOUNDATIONS AND LOGIC.

BERTRAND RUSSELL cables from London to the author:

"Your work is impressive and your erudition extraordinary. Have not had time for thorough reading but think well of parts read. Undoubtedly your theories demand serious consideration."

11. MATHEMATICAL PHYSICS.

B. F. DOSTAL, Professor of Mathematics, University of Florida.

"We still teach classical science on Mondays, Wednesdays and Fridays, and modern science on Tuesdays, Thursdays and Saturdays, as Sir William Bragg truly said. Within the bounds of the aristotelian system there appears to be no hope of ever finding the requisite unifying principle. Mathematicians have been rapidly outgrowing the old forms of so-called logic, but mathematical physicists have in general been slow to appreciate the value of these efforts or to apply these results to their own problems. Korzybski's *Science and Sanity* will be of great value to science because it contains the basis for the development of a new and wider, and more unifying, form of scientific determinism, without which the outlook for modern science would be gloomy indeed. Not only does Korzybski point out a more satisfactory, non-aristotelian, non-identity basis for a new science in general, than any hitherto employed, but he goes further in giving several promising suggestions for extensive developments and applications of the results of *modern* science, including those of the new wave and quantum mechanics. His work is bound to become a stimulus to investigators in mathematics, physics, chemistry, biology, 'psychology,' and medicine; and to educators, economists, sociologists, engineers, lawyers, and laymen as well, the majority of which still have a 'philosophy of the universe which takes one form on weekdays and another form on Sundays.'"

12. NEUROLOGY.

C. JUDSON HERRICK, Professor of Neurology, University of Chicago.

"The disturbances of mental balance and social stability now so prevalent seem to indicate a general failure to adjust our minds to our jobs. This results in futile conflict and too often in mental and social derangement. The numberless panaceas proposed fail because each attacks a single phase of a very complex situation, and generally a particular symptom rather than the cause of the trouble. Count Korzybski has diagnosed a fundamental source of confusion in thinking and in conduct and he presents a plan for radical revamping of our theory and practice that seems worthy of further trial in a wide variety of fields. His dynamic definition of *structure* in terms of relations gives promise of important applications in both science and practical affairs. It provides a generally useful symbol for experience of all sorts and a technique for recasting traditional ideas and practices more efficiently. Adjustments in terms of one dominant motive (or value) are replaced by a broader (many-valued) scheme of motivation which points the way toward personal and social sanity—a way that I believe is fundamentally correct and practicable."

See also C. M. CHILD, R. S. LILLIE, M. TRAMER, W. M. WHEELER, H. B. WILLIAMS, W. H. WILMER.

13. PHYSICS.

P. W. BRIDGMAN, Professor of Physics, Harvard University.

"Of late years the realization has been growing that the ultimate source of a large fraction of the difficulties of society, civilization, and science, is verbal in char-

acter. Among the few serious attempts to waken full self consciousness of what the situation is, and, having awaked consciousness, to provide *a technique* by which the vicious consequences of verbal habits may be avoided, I believe that of Count Korzybski must be rated as of the very first importance. I have been acquainted with his work for a number of years; not only do I believe it to be fundamentally sound, but I have always found his points of view most suggestive and stimulating, both in general and technical matters, and I have been amazed at the breadth of his interests and reading, and the diversity of the fields to which applications are made."

ROY J. KENNEDY, Professor of Physics, University of Washington, Seattle, Washington.

"Many of the impasses in which we of this lunatic world are involved are the result of verbal difficulties, and it is precisely these difficulties at which Count Korzybski's technique for the elimination of identity is chiefly aimed. He has shown a striking versatility in developing this technique which he has originated; he discusses the shortcomings of the sciences as facilely as those of religion. Whether or not the reader's sanity is improved by a careful study of the book, he cannot fail to enlarge his capacity for clear thinking. Paradoxically, athough *Science and Sanity* deals largely with the unspeakable it is suitable for discussion in the most decorous circles."

See also B. F. DOSTAL.

14. PHYSIOLOGY.

RALPH S. LILLIE, Professor of Physiology, University of Chicago.

"Count Korzybski's criticism of the present structure and usages of human society—as failing to keep pace with the advance of knowledge in the physical and biological sciences—is timely and well-founded, and is expressed with clearness, vigor and insight in this interesting book. It is certain that this knowledge, if widely diffused and acted upon, would greatly alleviate and perhaps remove many of the ills which afflict the modern world. The chief obstacle to such progress is not the lack of available knowledge, but the anachronistic survival of many mental habits and conceptions which are inconsistent with the facts of natural reality as revealed by science. These conceptions are firmly rooted in the general mind by language and custom. What is needed is a far-reaching revision of concepts, and this book points the way to such a revision. Since we are compelled by the conditions of existence to think and act in terms of symbols—concepts, words, images, formulae—it is all-important that these should conform as closely as possible to the permanent realities of life and nature. How to secure an adequate degree of such conformity and establish it by training and education is one of the most pressing problems of the time. Count Korzybski describes in detail the nature of verbal, mathematical and scientific symbolisms, and discusses clearly the biological, neurological, and other conditions which give them their representative value. He shows that misconceptions regarding the nature of language underlie many prevalent confusions and fallacies, especially the various fallacies of identification (arising mainly from verbalisms),—as when it is assumed that the application of the same label to different facts somehow renders them all alike and justifies the same action toward all. Serious consequences inevitably arise from the failures of discrimination and valuation thus resulting; and the author makes a special plea for discernment and individual treatment in the problems of human personality. It is only on this basis that many types of maladjustment can be prevented or corrected. These are only a few features in a book remarkable for its comprehensiveness, scholarship and independence."

HORATIO B. WILLIAMS, Dalton Professor of Physiology, Columbia University, and a Mathematician.

"In his *Science and Sanity* Count Korzybski undertakes to bring to the attention of his readers the importance of 'consciousness of abstracting,' or the permanent and full awareness that: (1) the object *is not* the event or the physico-chemical submicroscopic process; (2) that the symbol or label *is not* the object; and, (3) that an inference *is not* a description. Thus he is led to the formulation of a non-aristotelian system based on the complete rejection of 'identity.' To facilitate training

in non-identity he proposes a simple structural diagram which he calls the *Structural Differential.* With its aid, the general *verbal* rejection of 'identity,' is translated into *ordering* which becomes a visual, kinesthetic, *neuro*-psycho-logical method to train in non-identity or discrimination, and so to eliminate the always dangerous identifications, which play such an important rôle in all maladjustments.

It has been my privilege to read a considerable part of this book in manuscript and most of it in page proof. It appears at a time when the *modus vivendi* has broken down. National and international problems demand sane thinking. Count Korzybski, by rejecting a principle invariably false to facts (identity), which in principle prevents adjustment, points the way toward better adjustment, and so perhaps toward a saner solution of our man-made difficulties.

This is distinctly not a book for superficial reading, but one that will amply repay the thoughtful reader. It should be approached not as a medium of entertainment, though it may be that, but rather in the spirit indicated by one of its passages which refers to 'the joint labors of the author and the reader.' "

See also W. H. GANTT.

15. PSYCHIATRY.

DOCTOR PHILIP S. GRAVEN, Psychiatrist, Washington, D. C.

"I have read *Science and Sanity* through completely, some parts several times, and I must admit I have never encountered a work so rich in fundamental suggestions. It clearly covers a field almost wholly neglected in our University education. From the methodological point of view, therefore, the book is indispensable to any one endeavoring to carry on sane, clear, scientific work. This of course includes Medicine and especially Medical Psychology where sane thinking about the un-sane and 'insane' is vital at all times. Statements, principles, etc., that affect one's mental attitude toward problems are of most importance. These, *Science and Sanity* provides in abundance.

In addition to the scientists being considerably aided by the use of the non-aristotelian principles (aided in carrying on sane, creative, well-tempered, rigorous thinking about their observations and experimental data), there is also another group directly affected: namely, the mentally disordered. By direct clinical application, I have found the non-aristotelian principles workable in this enormous group. My observations cover a period of about six years. I shall have a great deal to say about these observations in contributions to medical and scientific journals.

By reading the book carefully, I have derived many benefits: personal, cultural, professional, scientific. The book appeals to me as one that will supply at least a few generations of scientific workers with a means of maintaining a productive, not clogging and obstructive psycho-logical attitude towards investigations needful to human security and advancement. And what could be more urgent in this modern age than a means to attain and maintain sanity: Korzybski's Theory of Sanity already makes that a possibility."

DOCTOR JOHN A. P. MILLET, Psychiatrist, New York City.

"It seems to me that Count Korzybski in his book *Science and Sanity* calls attention to one very important difficulty in the thinking process which he so aptly sums up as 'confusion in the orders of abstraction.' We have found through our analytical work in dealing with the neuroses that many difficulties in attitude and orientation toward life are derived from unconscious identification, a situation which leads to a real difficulty in individuation and sometimes makes it permanently impossible.

From the psychiatric point of view it seems to me that this one point that Korzybski has emphasized gives the chief value to his work. I doubt whether analysts, for the most part, have ever considered the problem of identification from exactly this point of view. Korzybski's presentation should be interesting to such a group and might well lead to further experimental activities in the field of education designed to offset the dangers of such identifications.

A beginning has been made in this field already by Korzybski in his development of the 'Structural Differential.' It is too early to say what practical value this Structural Diagram may have but the principles of its development are based on the conception of 'order' in neuro-psycho-logical processes and so provides what might

be described as a neuro-physiological technique for the elimination of false identification."

DOCTOR M. TRAMER, Priv.-Doz. in Psychiatry, University of Bern, Switzerland; Medical Director State Asylum, President Swiss Psychiatric Association. A Mathematician, Neurologist, formerly a co-worker with the late von Monakov, and a Psychiatrist.

"For almost three years I have had the opportunity to follow the researches of Count Korzybski. I recognized already from the formulation of 'time-binding' that his work deals with something fundamental. Further results of his researches justified this opinion. These affect not only our scientific thinking but also our daily lives. Korzybski now presents his work in a comprehensive form to the public, its study reveals that we have to deal with a structure of deeper foundations, the beginning of which have already appeared in different scientific domains, and particularly in physics. The general structure of the completed development of his thesis leads to the formulation of the urgent need for a fundamental analysis (revision) of our scientific language for which he blazes the trail. The means he selects to overcome the consequent difficulties of such an analysis, and to bring into the foreground the necessity for general structural thinking above all, are, in my opinion, results of fundamental importance."

DOCTOR WILLIAM A. WHITE, Professor of Psychiatry, George Washington University Medical School; Superintendent Saint Elizabeth's Hospital, Washington, D. C.

"Korzybski's concepts are to me very helpful, particularly his concepts of linearity, multiordinality, degrees of abstractions and finally of non-identity. I have always felt that such matters as are included in these concepts are of extreme importance to have as one's mental equipment. I am at one with Korzybski in many ways. Certainly I am sure that how we think about things is of as much importance as what we think about them. I congratulate the author upon the appearance of his book, and I think he has made a real contribution to the methodology of thinking."

16. SEMANTICS.

See B. MALINOWSKI.

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